WHY?

Module 1

Materials for Concrete: Cement, Manufacturing Process, Physical Properties, Hydration of Cement, hydration products, Chemical Compounds in Cement, Types of Cement, Aggregates: Classification of aggregates, Physical Properties, Bulking of Sand, Mechanical Properties, Water: Specifications of Water to be used For Concrete

Module 2

Properties of Fresh Concrete -Types of Batching, Mixing, Transportation, Placing Including Pumping and Compaction. Techniques for Good Quality Concrete, Workability, Factors affecting workability, Methods of Measuring Workability, Segregation and Bleeding, setting time, Curing of Concrete, Types of curing, Temperature Effects on Fresh Concrete

Module 3

Admixtures in Concrete: Types, Plasticizers and Super-plasticizers and their Effects On Workability, Air Entraining Agents, Accelerators, Retarders, Pozzolanic Admixtures, Green concrete, Bonding Admixtures, Damp-Proofing Admixtures, Construction Chemicals

Module 4

Desired Properties of Concrete, Strength, Durability & Im-permeability, Characteristic Strength, Compressive, Tensile and Flexure of Concrete, **Bond Strength,** Tests on Concrete, Modulus of Elasticity, Effect of W/C Ratio and admixtures on Strength, Types **of concrete,** High Strength and High Performance Concrete

Module 5

Creep and Shrinkage of Concrete, Significance, Types of Shrinkage and Their Control, Factors Affecting Creep. Durability of Concrete: Minimum & Maximum Cement Content, Strength & Durability Relationship, Exposure to Different Conditions, Factors Contributing to Cracks in Concrete, Sulphate Attack, Alkali Aggregate Reaction (AAR),factors affecting on AAR, Deteriorating effects of AAR, Chloride Attack, Corrosion of Steel (Chloride Induced)

Module 6

Concrete Mix Design, Nominal Mix Concrete, Factors Governing Mix Design, Methods Of Expressing Proportions, Trial Mixes, Acceptance Criteria, Factors Causing Variations, Field Control, Statistical Quality Control, Quality Measurement in Concrete Construction., Non-destructive Testing of Concrete

Text Books

- Gambhir M. L. " Concrete Technology" , Tata Mc-Graw Hill 2015 15th edition
- Shetty M. S. "Concrete Technology" , S. Chand 2005.
- Krishnaswamy, " Concrete Technology" , Dhanapat Rai and Sons

Reference Books

- Orchard, " Concrete Technology" , Applied Science Publishers
- Neville A. M., "Concrete Technology" , Pearson Education
- IS:10262(2009), IS:456 (2009), IS 4926 (2003)

- CEMENT
- MANUFACTURE OF CEMENT
- MIXING OF RAW MATERIALS
- WET PROCESS
- DRY PROCESS
- CHEMICAL COMPOSITION OF CEMENT
- BOGUE’S COMPOUNDS
- HYDRATION OF CEMENT
- CLASSIFICATION OF CEMENT
- TEST ON CEMENT
- AGGREGATE AND CLASSIFICATION OF AGGREGATES
- MECHANICAL PROPERTIES OF AGGREGATES
- PHYSICAL PROERTIES OF AGGREGATES
- SOUNDNESS OF AGGREGATES
- BULKING OF FINE AGGREGATES
- SIEVE ANALYSIS
- FINENESS MODULUS
- RECYCLED AGGREGATE
- WATER AND TEST ON WATER

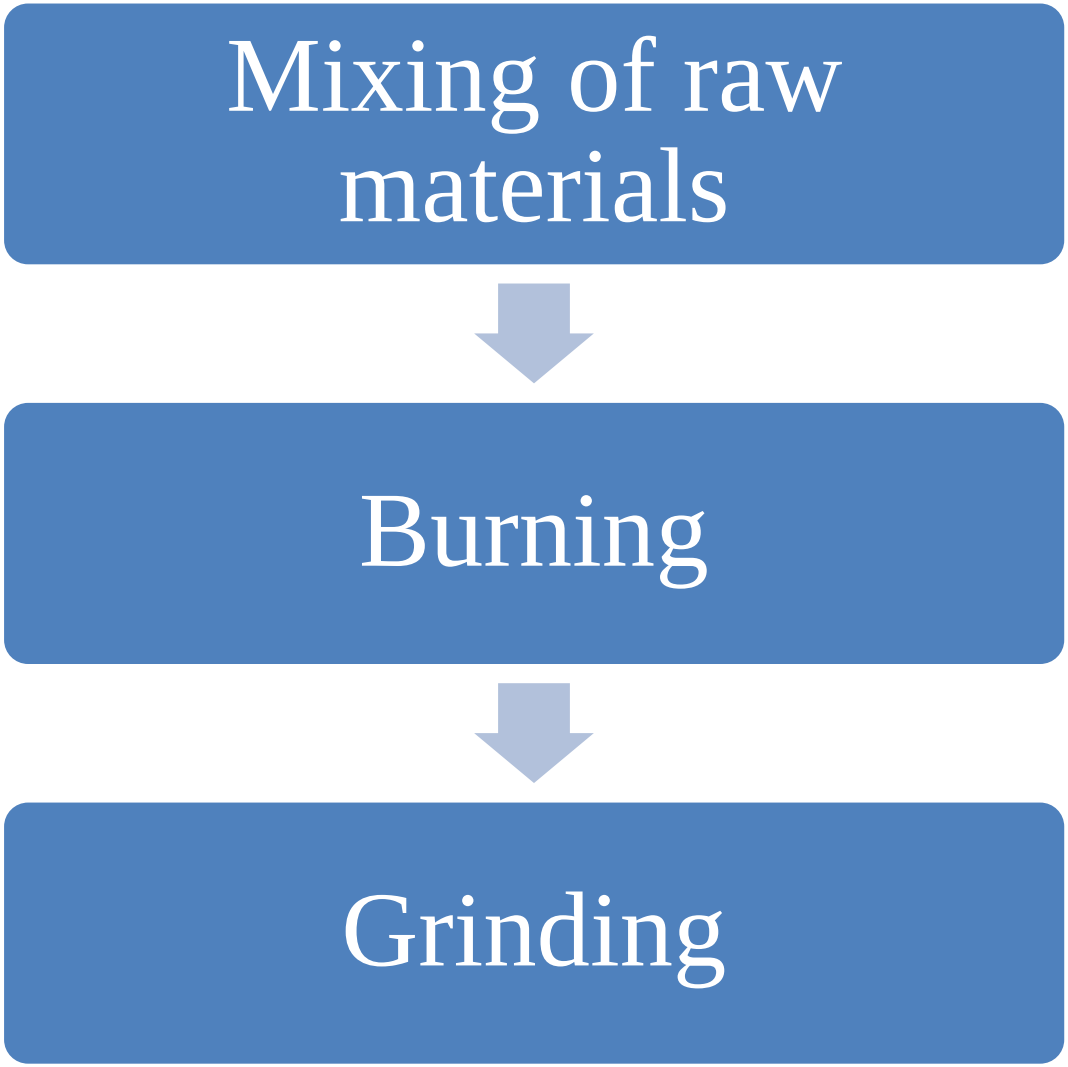
History

- 1756-John Smeaton (England) created hydraulic lime from lime stone.
- 1796- John Parker(England) created an improved version. Birth of natural cement industry.
- 1824-portland cement was invented by Joseph Aspedin. It resembled the color of sand stone found in Portland, U.K .
- 1904-First cement factory in Tamil Nadu, India.

Cement

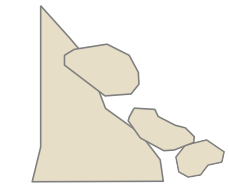
- Cement is the mixture of calcareous, siliceous, argillaceous and other substances. Cement is used as a binding material in mortar, concrete, etc.
- It was invented by Joseph Aspdin of U.K. in 1824
- Joseph Aspdin took out a patent in 1824 for "Portland Cement," a material he produced by firing finely-ground clay and limestone until the limestone was calcined. He called it Portland Cement because the concrete made from it looked like Portland stone, a widely-used building stone in England

Manufacture of Portland Cement

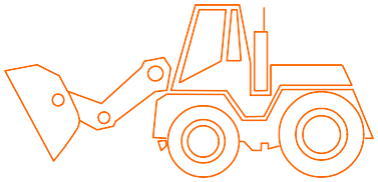


THE CEMENT MANUFACTURING PROCESS

quarry

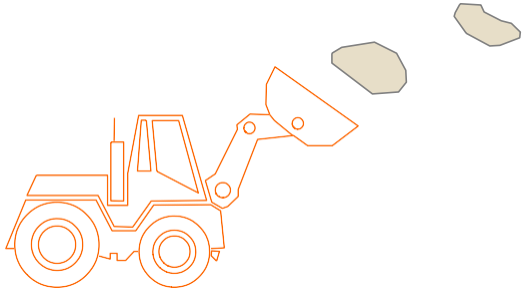


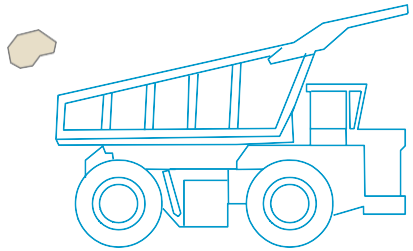
Quarry face



loader

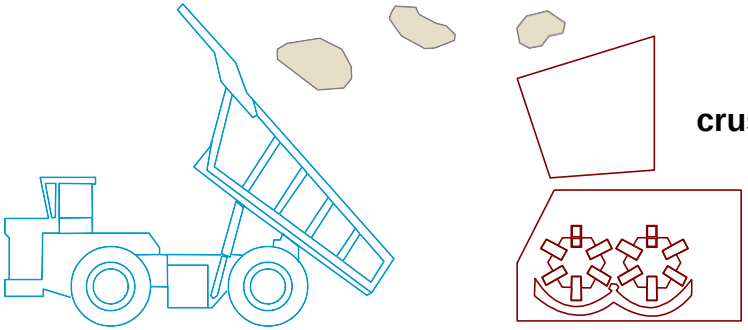
1. BLASTING





dumper

2. TRANSPORT

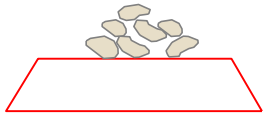


crushing



conveyor

storage at the plant



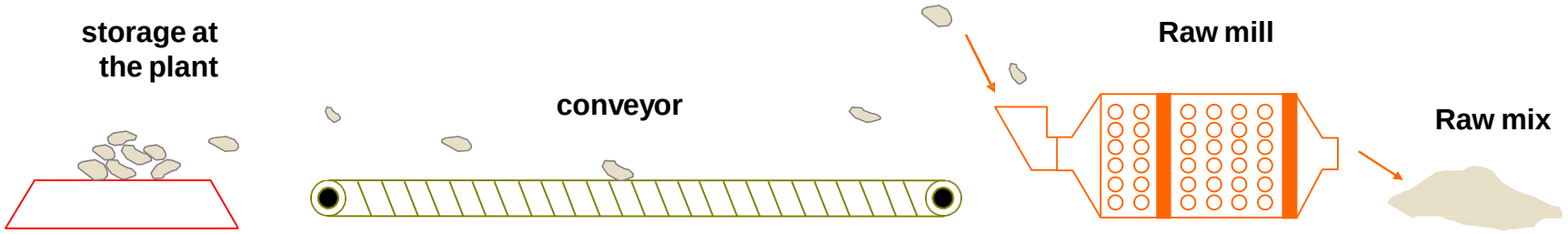
3. CRUSHING & TRANSPORTATION

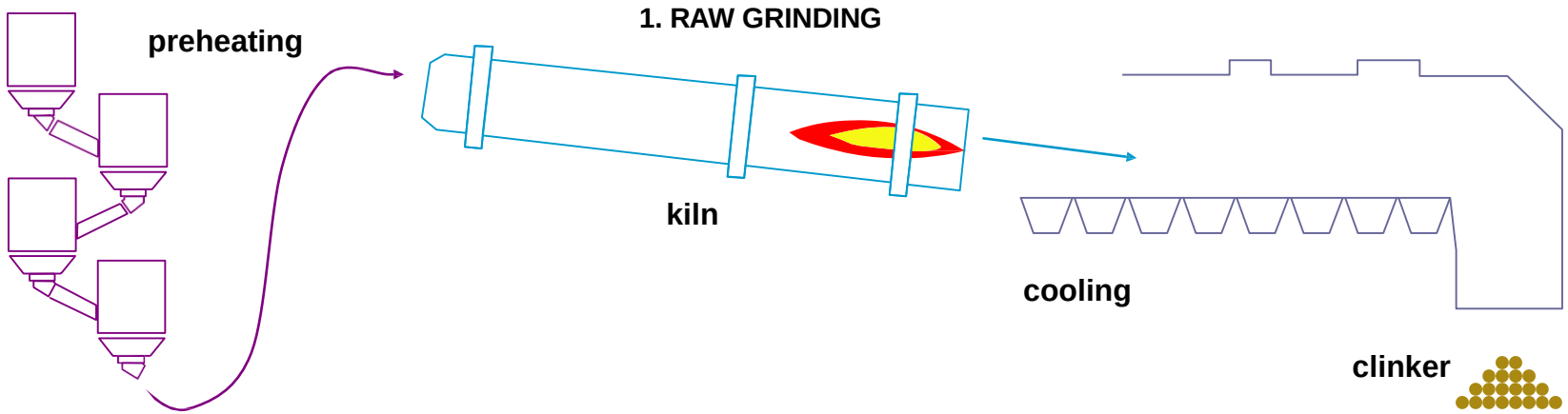
1.BLASTING : The raw materials that are used to manufacture cement (mainly limestone and clay) are blasted from the quarry.

2.TRANSPORT : The raw materials are loaded into a dumper.

3.CRUSHING AND TRANSPORTATION : The raw materials, after crushing, are transported to the plant by conveyor. The plant stores the materials before they are homogenized.

Raw grinding and burning



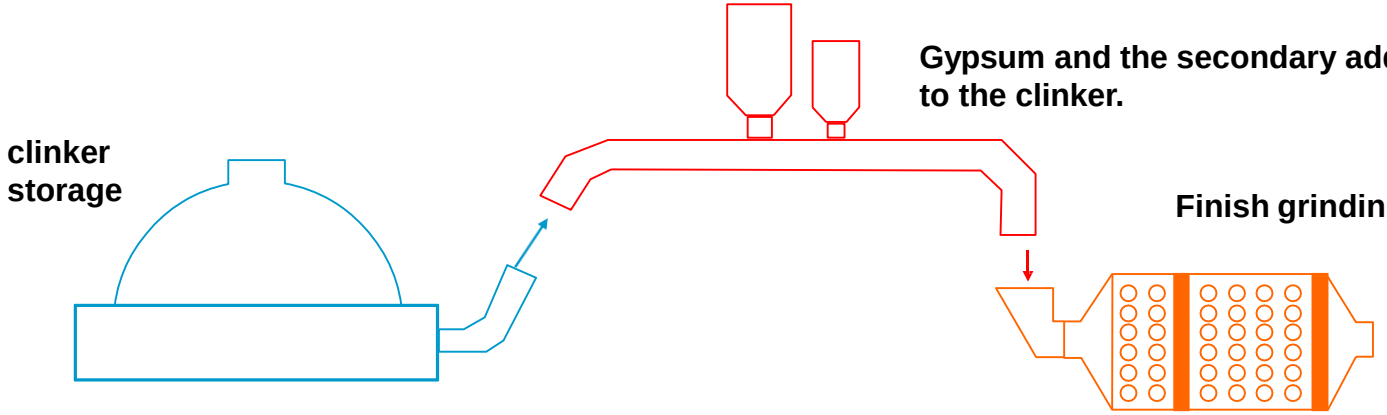


2. BURNING

1.RAW GRINDING : The raw materials are very finely ground in order to produce the raw mix.

2.BURNING : The raw mix is preheated before it goes into the kiln, which is heated by a flame that can be as hot as 2000 °C. The raw mix burns at 1500 °C producing clinker which, when it leaves the kiln, is rapidly cooled with air fans. So, the raw mix is burnt to produce clinker : the basic material needed to make cement.

Grinding, storage, packing, dispatch



1. GRINDING

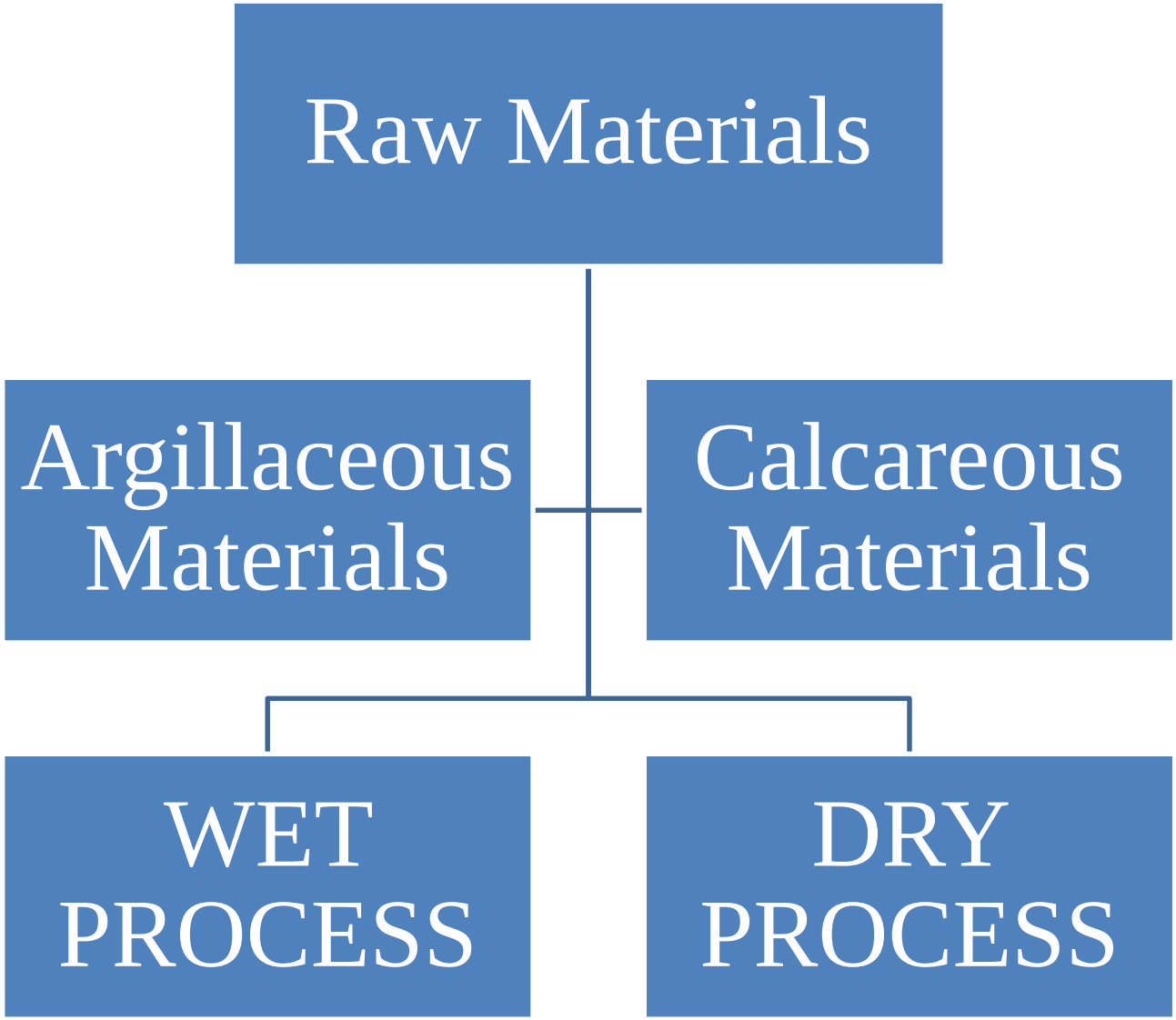


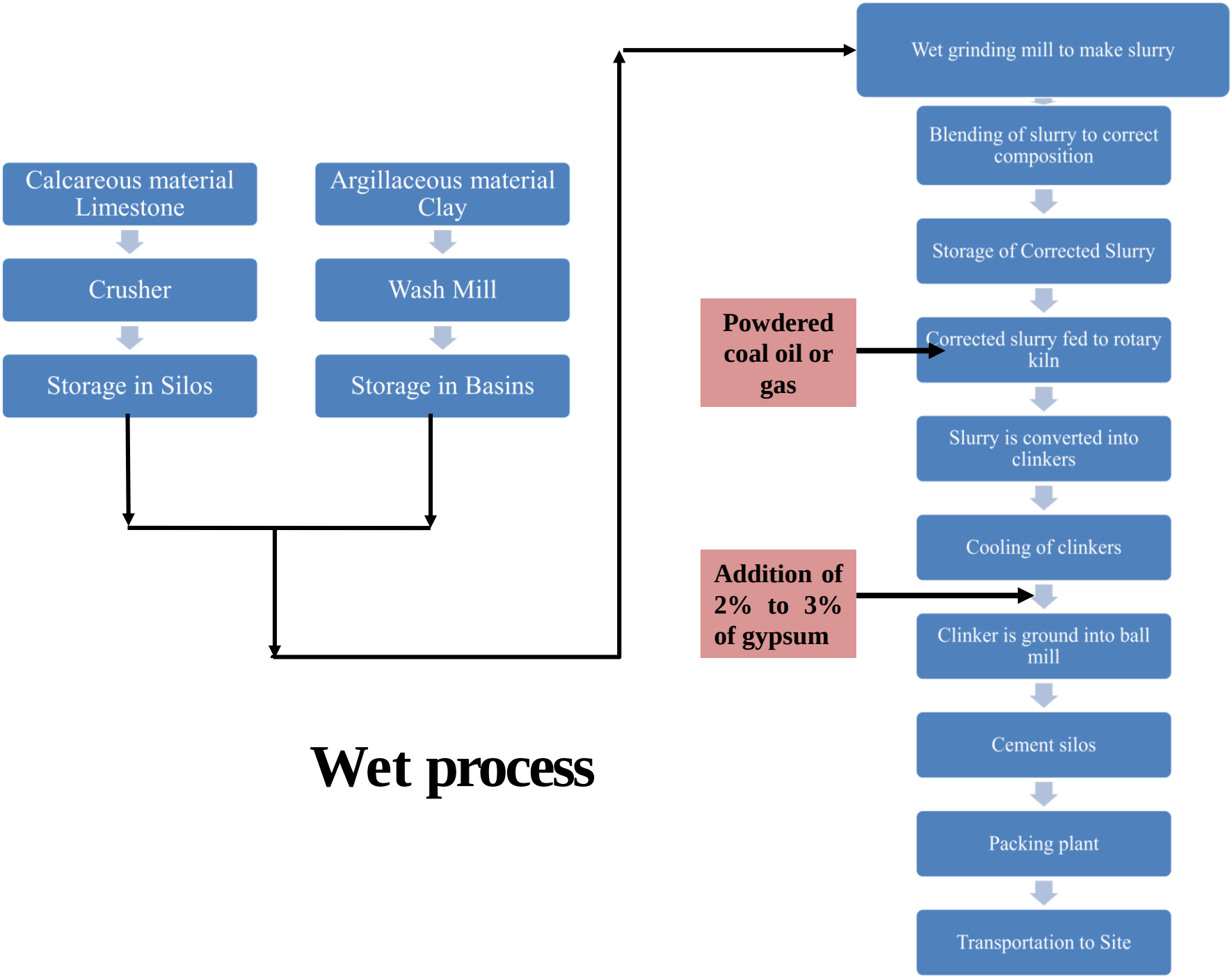
2. STORAGE, PACKING, DISPATCH

1.GRINDING : The clinker and the gypsum are very finely ground giving a “pure cement”. Other secondary additives and cementitious materials can also be added to make a blended cement.

2.STORAGE, PACKING, DISPATCH : The cement is stored in silos before being dispatched either in bulk or in bags to its final destination.

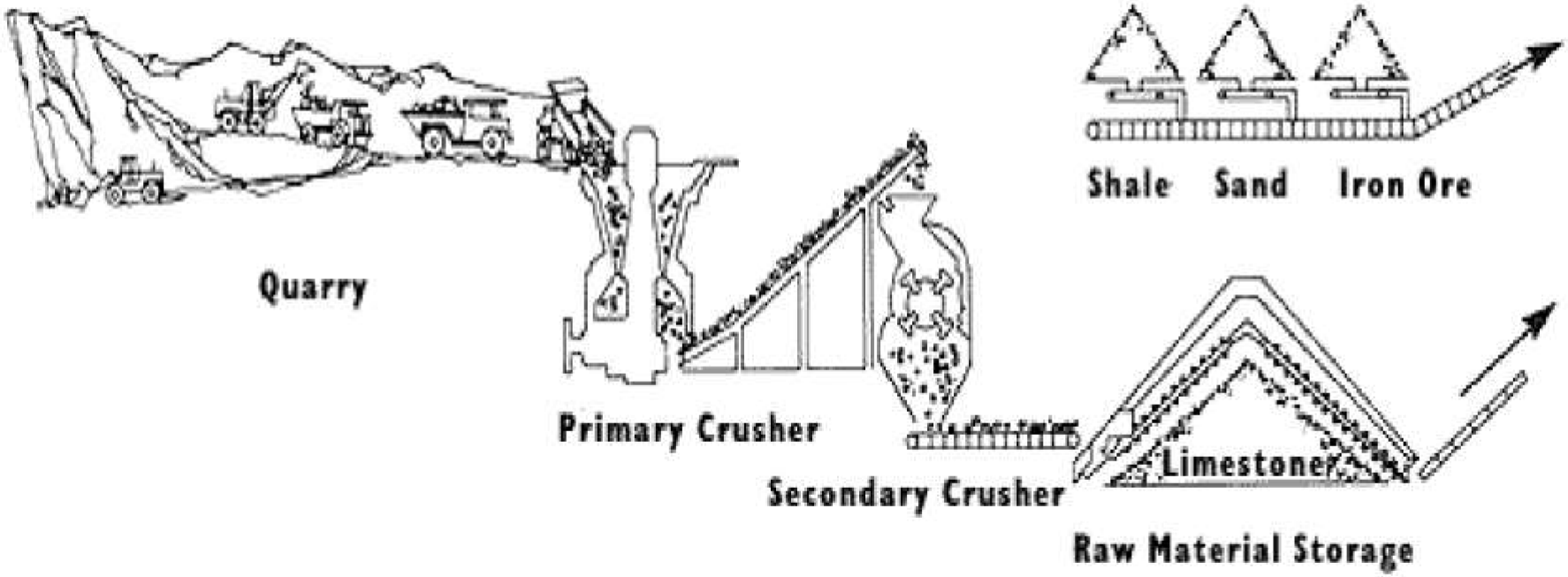
Mixing of Raw Materials




Calcareous material
Limestone
Crusher
Storage in Silos
Argillaceous material
Clay
Wash Mill
Storage in Basins
Wet grinding mill to make slurry
Blending of slurry to correct composition
Storage of Corrected Slurry
Powdered coal oil or gas
Corrected slurry fed to rotary kiln
Slurry is converted into clinkers
Cooling of clinkers
Addition of 2% to 3% of gypsum
Clinker is ground into ball mill
Cement silos
Packing plant
Transportation to Site

Wet process

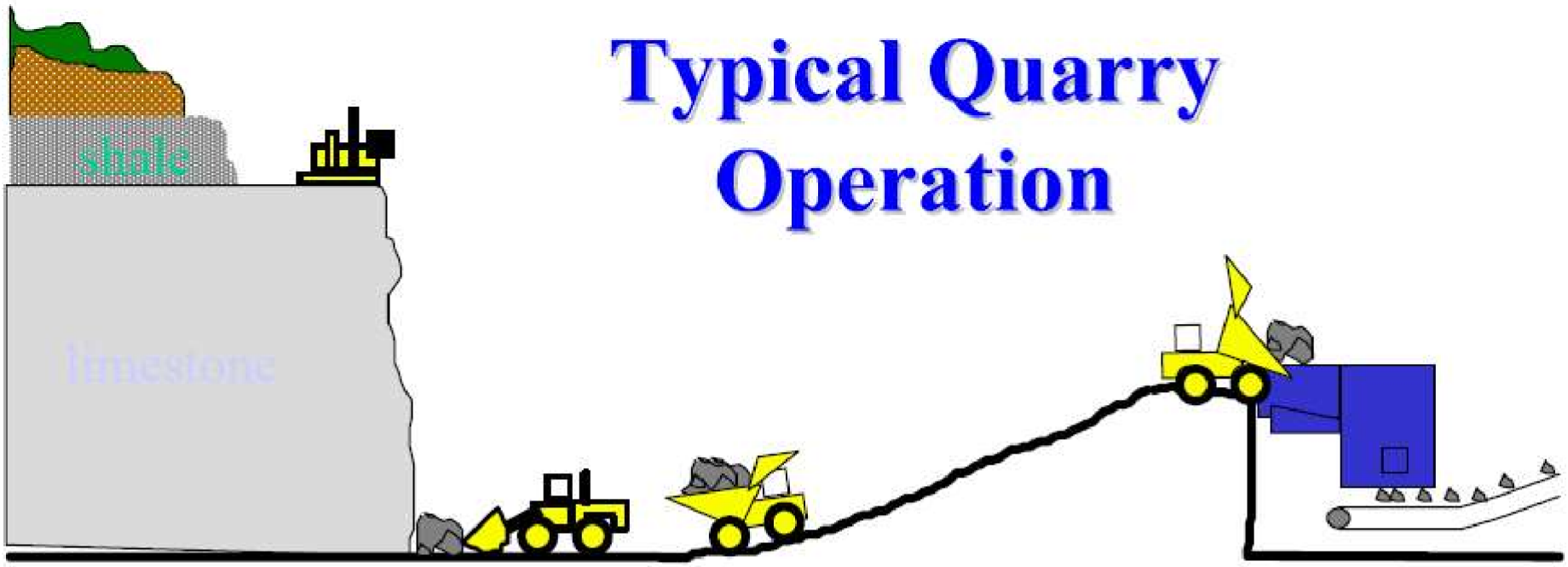
Steps in the Manufacture of Portland Cement



Rock mined from a quarry is crushed in either one or two stages, then stored with other raw materials to await further processing.

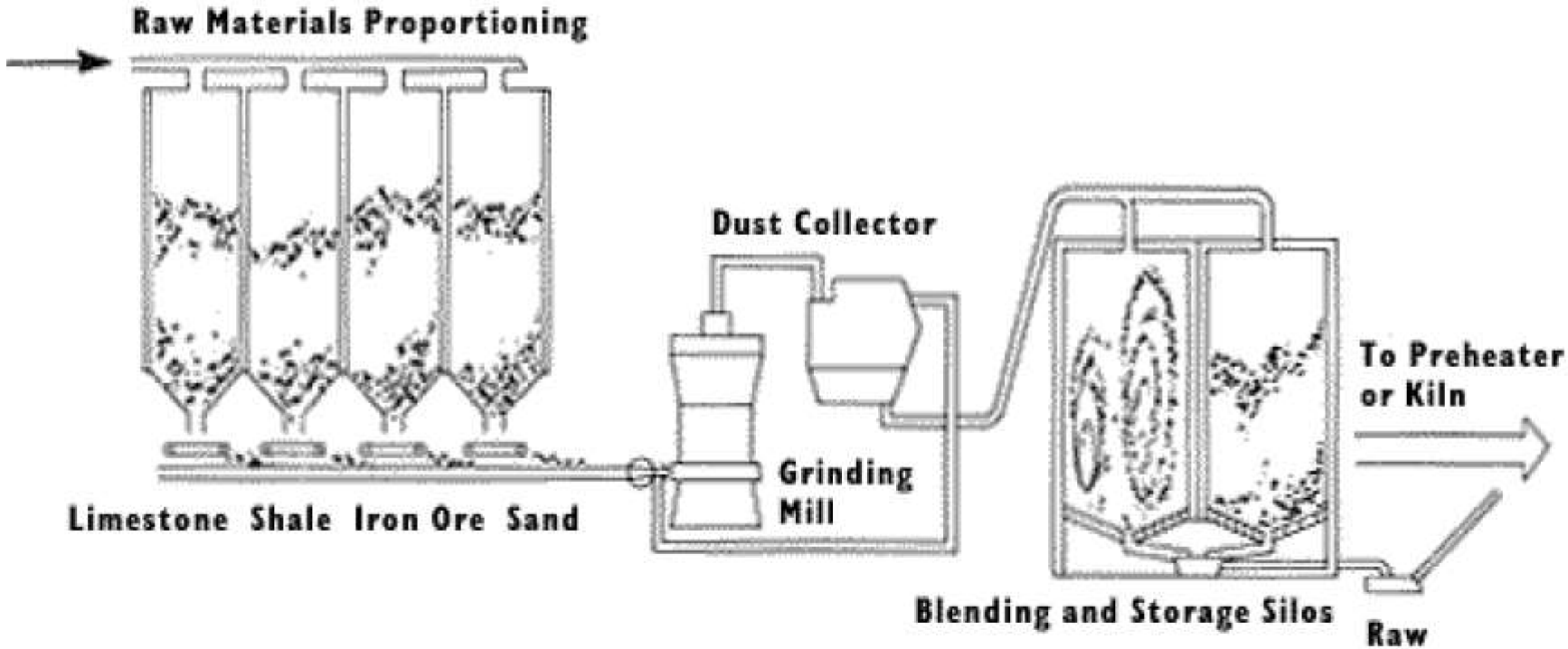
To Proportioning and Grinding

Typical Quarry Operation



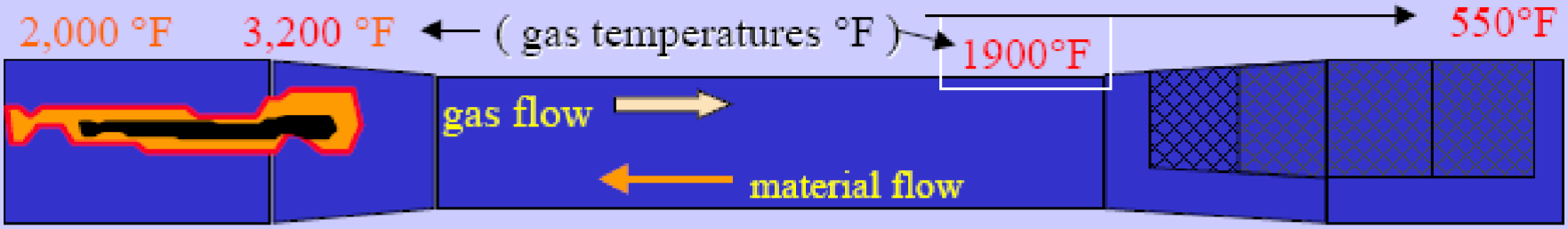
- Typically shale provides the argillaceous components:
  - Silica (SiO_2, Aluminum(Al_2O_3) & Iron (Fe_2O_3)
- Limestone provides the calcareous component:
  - Calcium Carbonate ($CaCO_3$)
- Raw materials may vary in both composition and morphology.

Steps in the Manufacture of Portland Cement



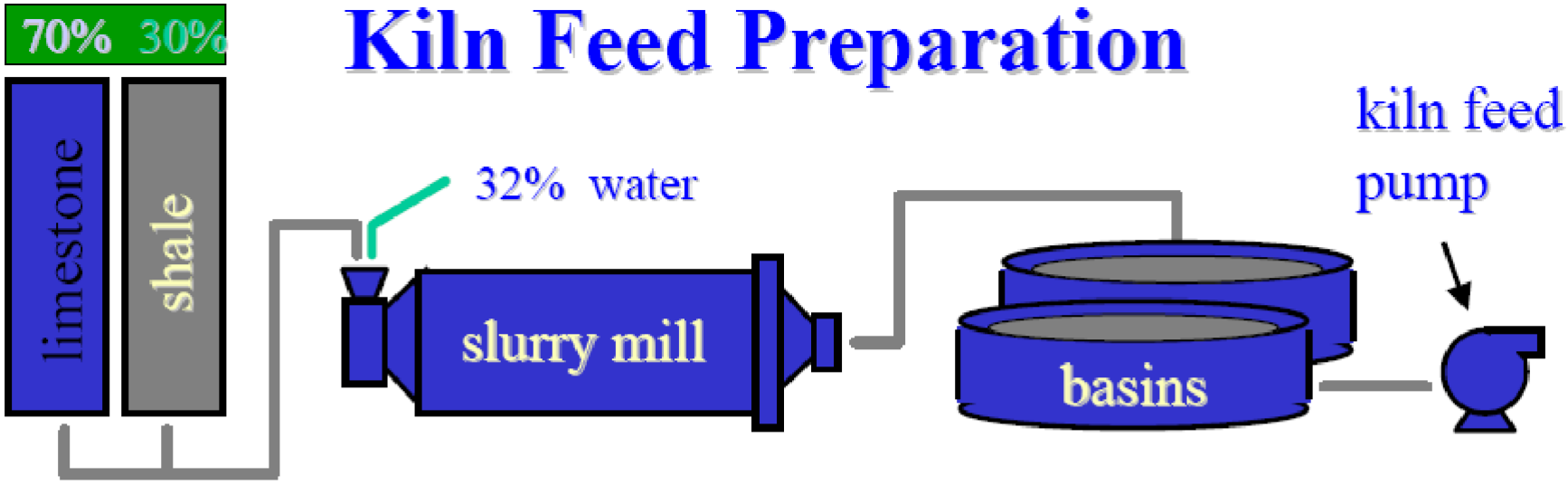
After analysis, the raw materials are proportioned, ground to fine powder, and blended. Some cement plants add water to the material during grinding, then blend and store it as a slurry.

The Wet Cement Kiln

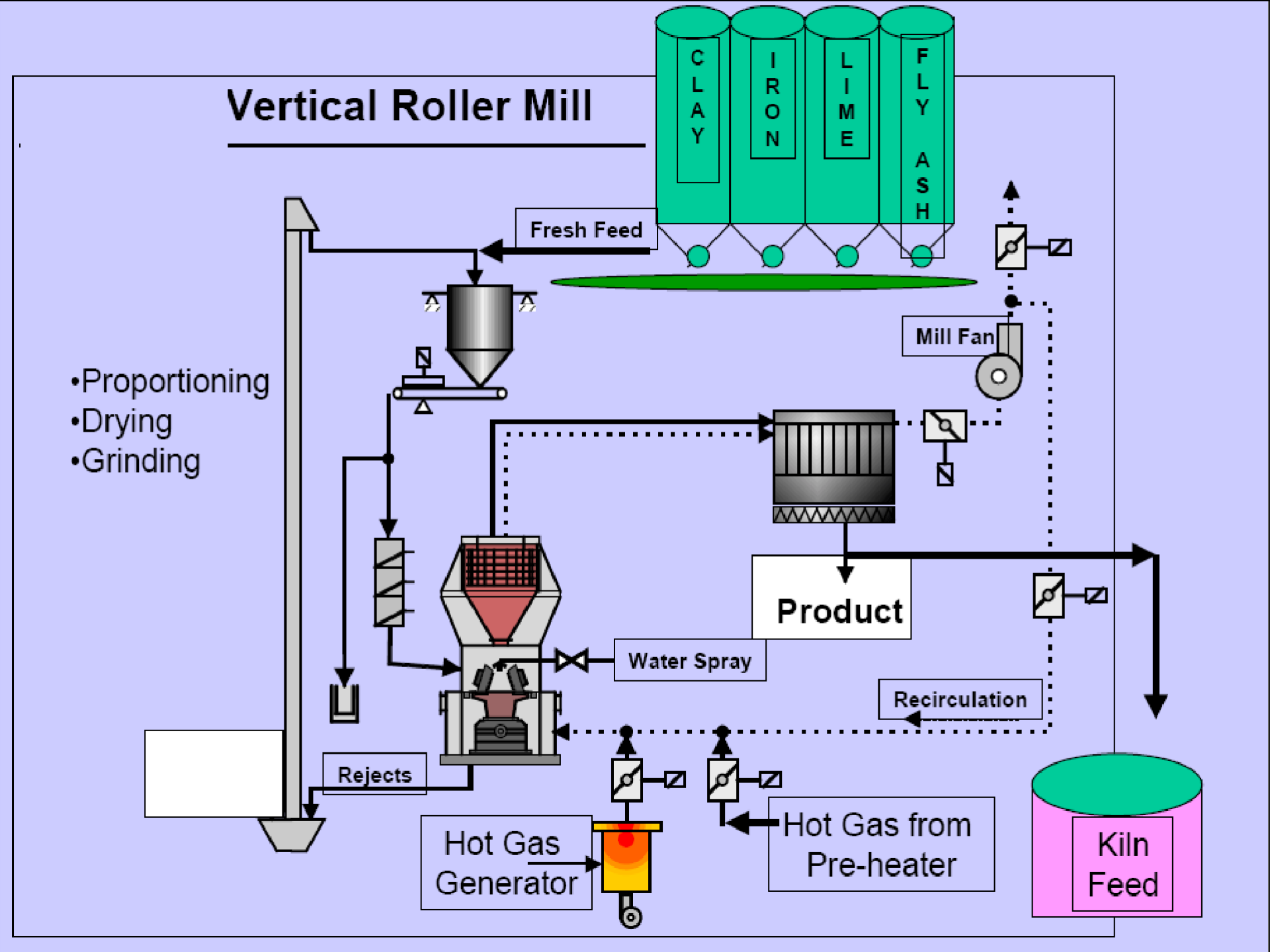


- The kiln is a continuous stream process vessel in which feed and fuel are held in dynamic balance
- 5 distinct process functions are performed in the kiln:
  - Dry
  - Preheat
  - Calcine
  - Sinter
  - Cool

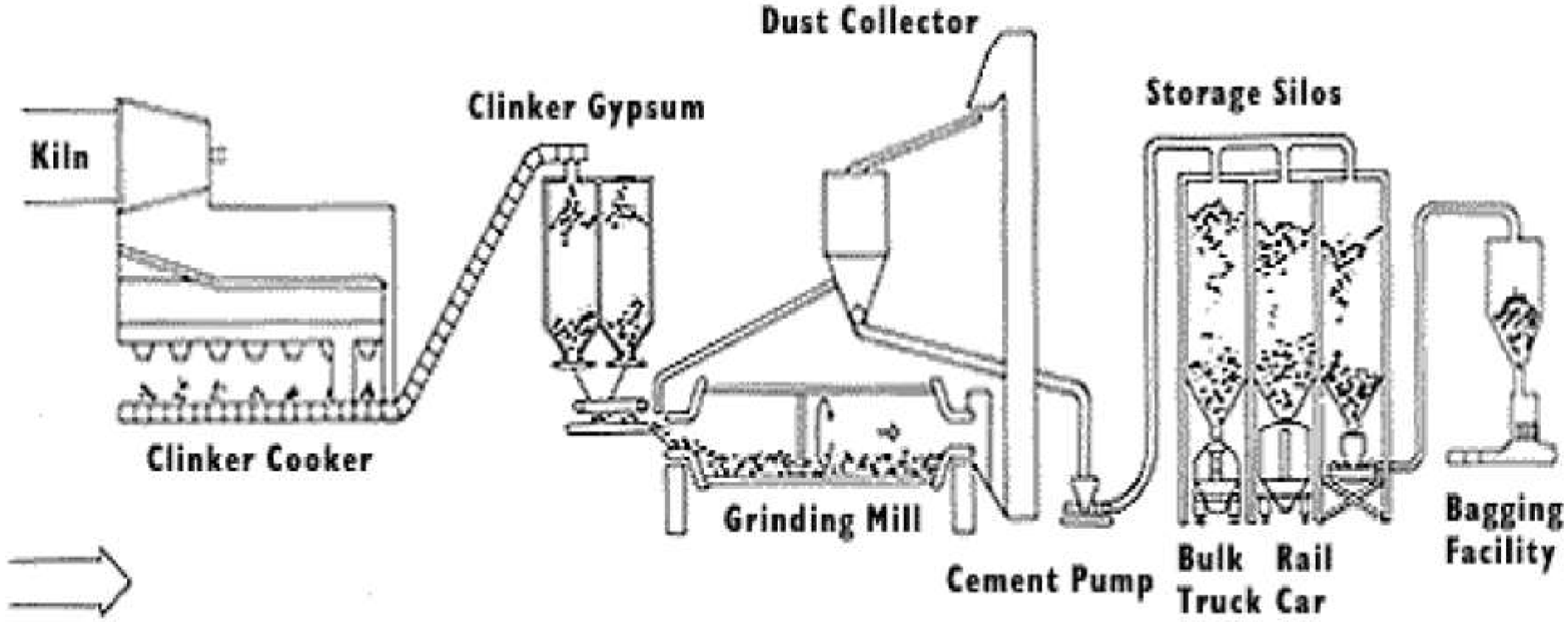
Kiln Feed Preparation



- Proportioning of feed stock.
- Size reduction to < 125μ.
- Control of moisture.
- Blending to reduce standard deviation.
- Uniform delivery rate of feed to the Kiln.

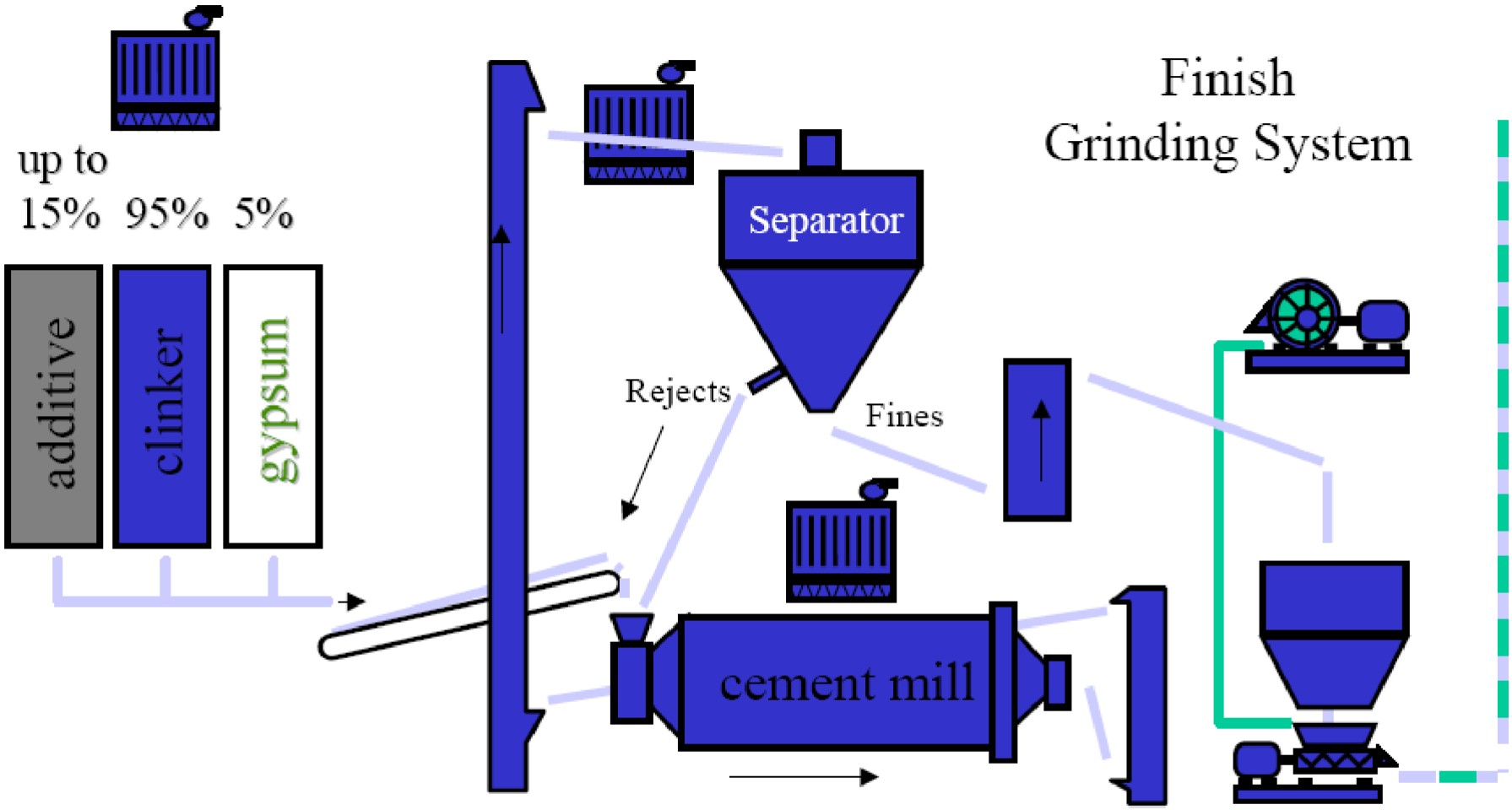

Vertical Roller Mill
CLAY
IRON
LIME
FLY ASH
Fresh Feed
Mill Fan
•Proportioning
•Drying
•Grinding
Product
Water Spray
Recirculation
Rejects
Hot Gas Generator
Hot Gas from Pre-heater
Kiln Feed

Steps in the Manufacture of Portland Cement

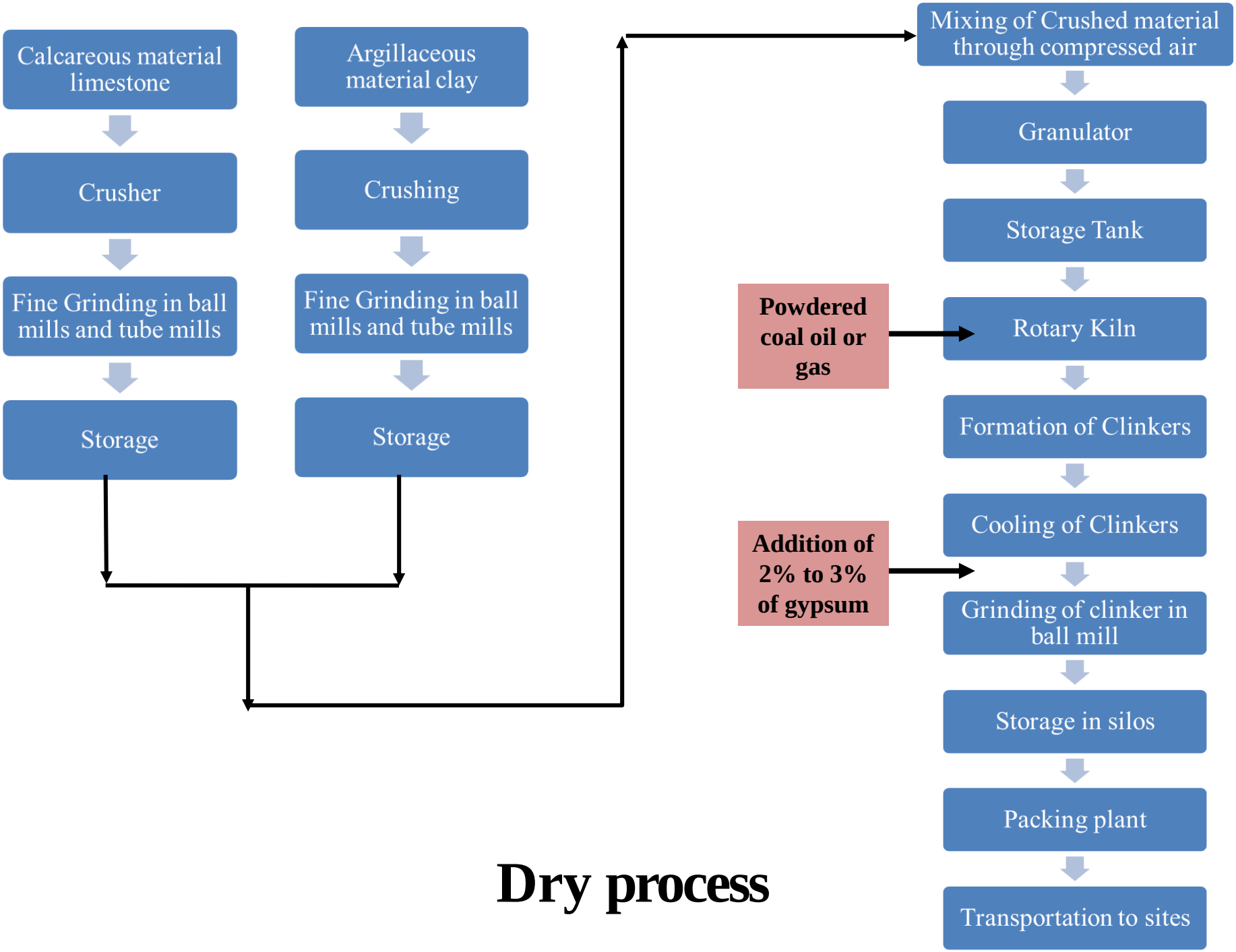


Once cooled, the clinker is ground with a small amount of gypsum.
It's now portland cement-ready to be bagged or shipped in bulk.

Finish Grinding System



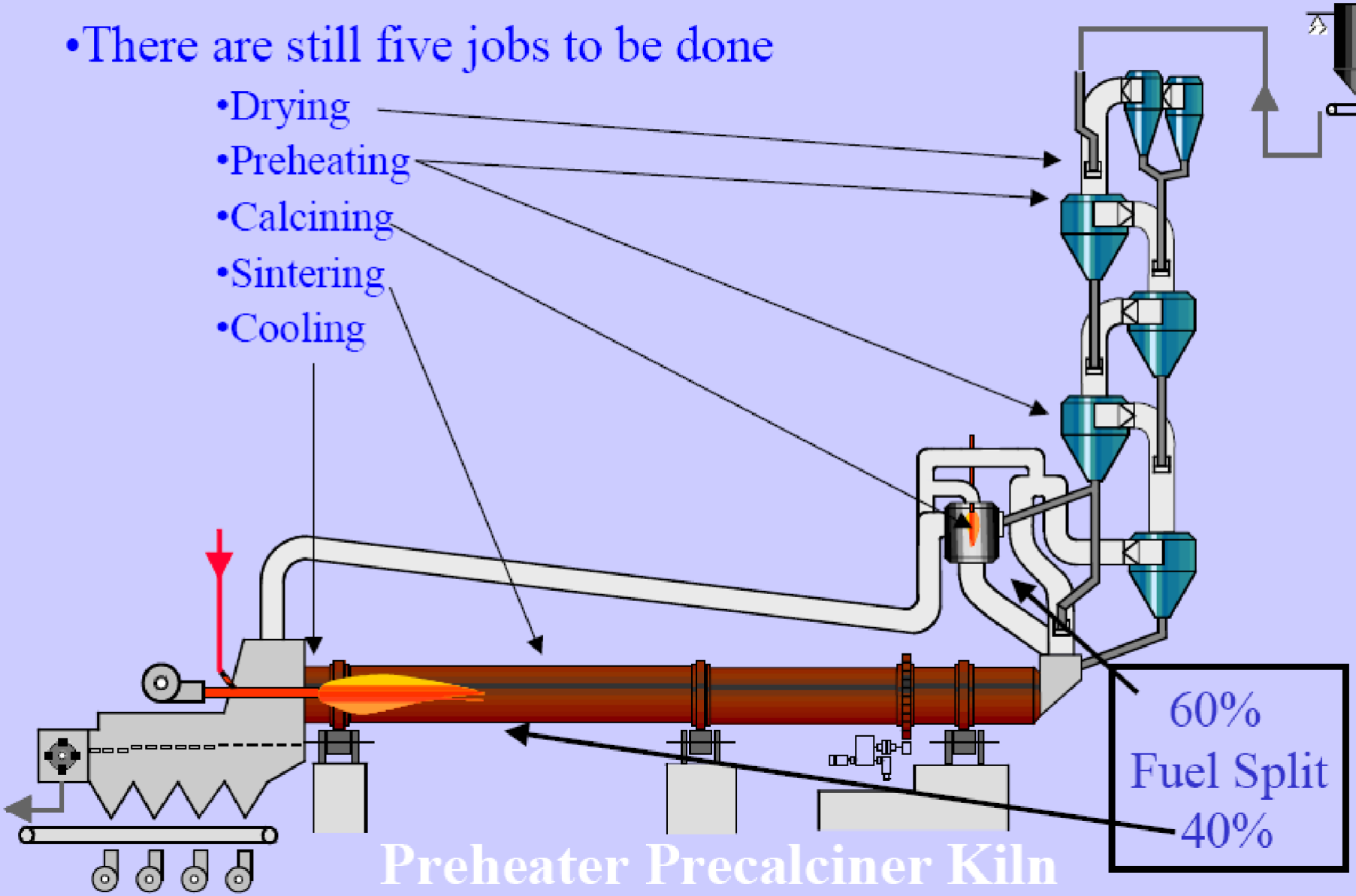
- Clinker, gypsum, and optional additives are weighed to proper proportions and ground in the cement mills.
- Additives may include: Fly-ash, Limestone…..

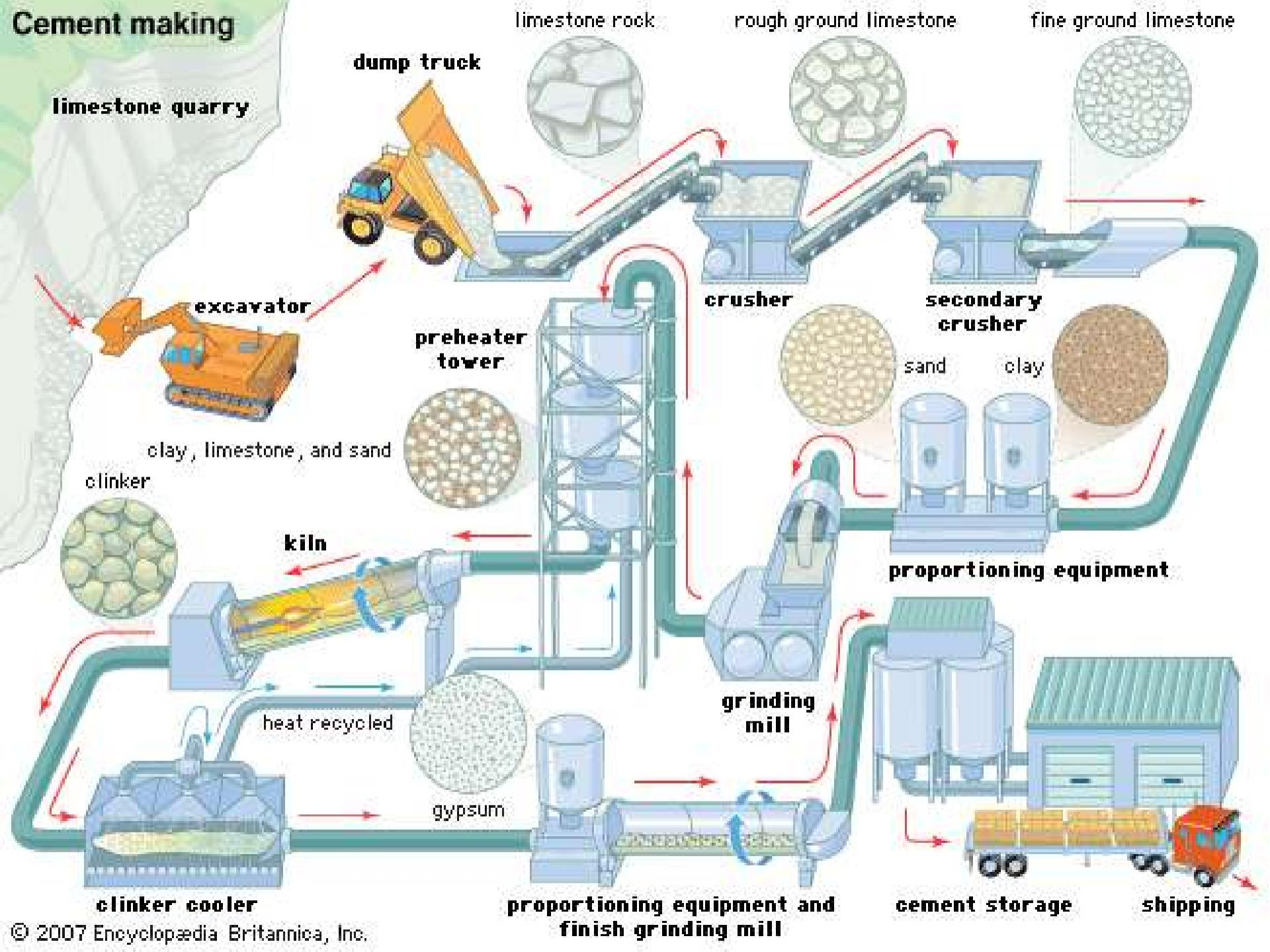

Calcareous material limestone
Crusher
Fine Grinding in ball mills and tube mills
Storage
Argillaceous material clay
Crushing
Fine Grinding in ball mills and tube mills
Storage
Mixing of Crushed material through compressed air
Granulator
Storage Tank
Powdered coal oil or gas
Rotary Kiln
Formation of Clinkers
Cooling of Clinkers
Addition of 2% to 3% of gypsum
Grinding of clinker in ball mill
Storage in silos
Packing plant
Transportation to sites

Dry process

Dry Process Preheater/Precalciner System

- There are still five jobs to be done
  - Drying
  - Preheating
  - Calcining
  - Sintering
  - Cooling





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Chemical Composition of O.P.Cement:

O.P.C has the following approximate chemical composition:

The major constituents are:

1. Lime	(CaO)	60- 63%
2. Silica	(SiO2)	17- 25%
3. Alumina	(Al2O3)	03- 08%

Chemical Composition of O.P.Cement: Continued-------

The auxiliary constituents are:

1. Iron oxide (Fe2O3) 0.5- 06%
2. Magnesia (MgO) 1.5- 03%
3. Sulphur Tri Oxide (SO3) 01- 02%
4. Gypsum 01 to 04%

(i) Lime (CaO):

1.Lime forms nearly two-third (2/3) of the cement. Therefore sufficient quantity of the lime must be in the raw materials for the manufacturing of cement.

2.Its proportion has an important effect on the cement. Sufficient quantity of lime forms di-calcium silicate and tri-calcium silicate in the manufacturing of cement.

3.Lime in excess, causes the cement to expand and disintegrate.

(ii) Silica (SiO_2):

1.The quantity of silica should be enough to form di-calcium silicate and tri-calcium silicate in the manufacturing of cement.

2. Silica gives strength to the cement.

3.Silica in excess causes the cement to set slowly.

(iii) Alumina (Al_2O_3):

1. Alumina supports to set quickly to the cement.

2. Lowers the clinkering temperature.

3. Alumina in excess, reduces the strength of the cement.

(iv) Iron Oxide (Fe_2O_3):

Iron oxide gives colour to the cement.

(v) Magnesia (MgO):

1.It also helps in giving colour to the cement.
2.Magnesium in excess makes the cement unsound.

(vi) <u>Calcium Sulphate (or) Gypsum (CaSO4):</u>

At the final stage of manufacturing, gypsum is added to increase the setting of cement.

Bogue's Compounds

C3S & C2S PRVIDE STRENTH & OFFERS RESISTANCE TOWARDS THE

Name of Compound	Formula	Abbreviated Formula
Tricalcium silicate	3 CaO.SiO2	C3S –hydrates rapidly; provides early & ultimate strength
Dicalcium silicate	2 CaO.SiO2	C2S-hydraes slowly& provides strength after a duration of 7 days
Tricalcium aluminate	3 CaO.Al2O3	C3A-hydrates rapidly; provides early strength & less ultimate strength
Tetracalcium aluminoferrite	4 CaO.Al2O3.Fe2O3	C4AF-doesn't provide strength but is stable than C3A

Chemical Composition of PC

- **Calcination in the kiln changes molecular structure of ingredients.**

Main compounds:

- tricalcium silicate (C_3S)
- dicalcium silicate (C_2S)
- tricalcium aluminate (C_3A)
- tetracalcium aluminoferrite (C_4AF)

Minor compounds: small percent but can have strong influences:

magnesium oxide
titanium oxide
manganese oxide
sodium oxide
potassium oxide

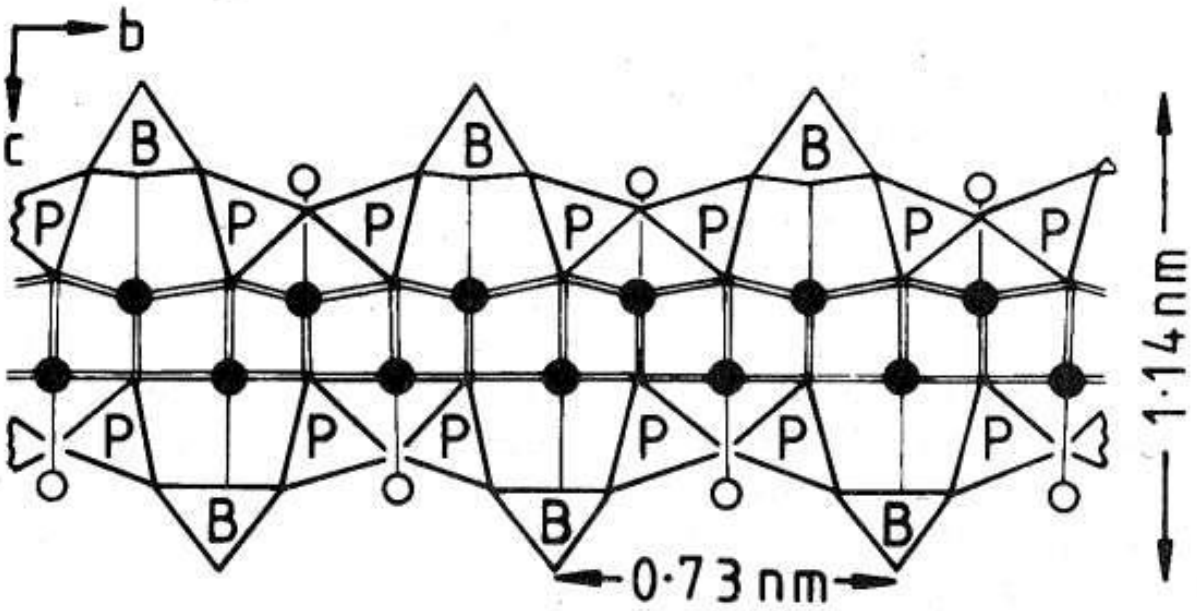
Alkalis (Na_2O, K_2O) react with silica causing disintegration & expansion of concrete

HYDRATION- INTRODUCTION

Chemical reaction take place between cement and water is referred as hydration.

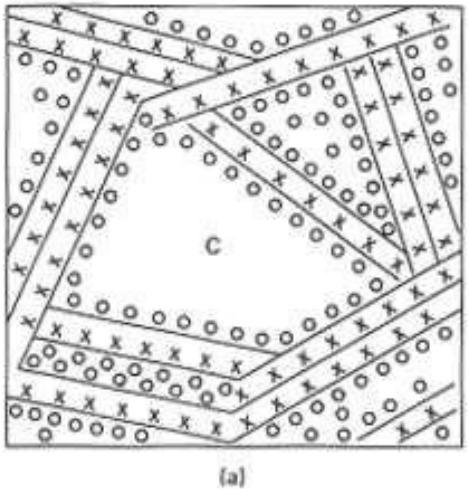
Hydration is the reaction(Series of chemical reactions) of cement with water to form the binding material(C-S-H Gel).

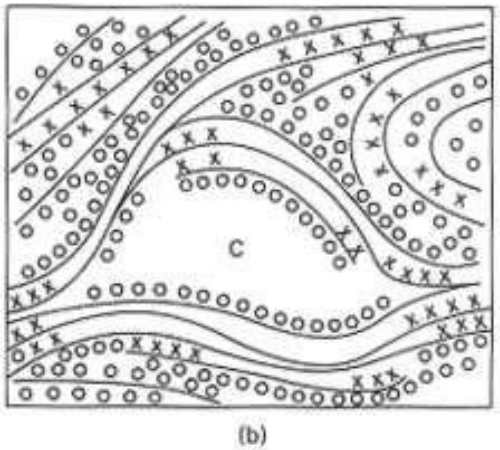
Structure of C-S-H gel



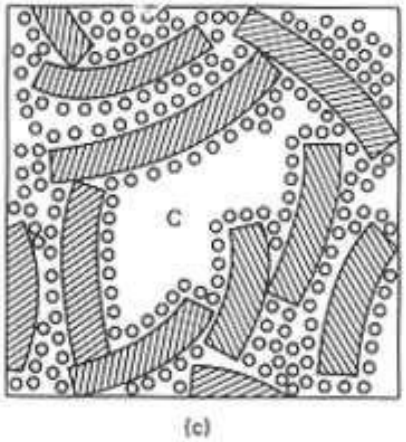
- Structure of 1.4 nm tobermorite, a sheet like silicate composed of octahedral layers and silicate chains.
- The silica tetrahedra can be replaced by hydroxil ions.
- If part the bridging tetrahedra (B) are replaced only paired groups remain explaining the dimer signal in NMR studies.

Structure of C-S-H gel





Feldman and Sereda (1970)



Munich model, Wittman (1979)

Colloid model

C-S-H gel models

x Structural water
Adsorbed water
c Capillary pore
C-S-H layer
C-S-H particle

HYDRATION- INTRODUCTION

Bogue's Compounds

Name of the Compound	Formula	Abbreviated Formula	Percentage
Tri-calcium Silicate	$3CaO.SiO_2$	C_3S	25-50
Di-calcium Silicate	$2CaO.SiO_2$	C_2S	20-45
Tri-Calcium Aluminate	$3CaO.Al_2O_3$	C_3A	5-12
Tetra Calcium Aluminoferrite	$4CaO.Al_2O_3.Fe_2O_3$	C_4AF	6-12

HYDRATION-INTRODUCTION

In presence of water, the silicates(C_3S and C_2S) and Aluminates(C_3A and C_4AF) form products of hydration which in time produce a firm(Solid) and hard mass - **the hydrated cement paste**

HYDRATION PROCESS - CALCIUM SILICATE (C_3S & C_2S)

HYDRATION PROCESS-C_3S

Calcium Silicate Hydrates

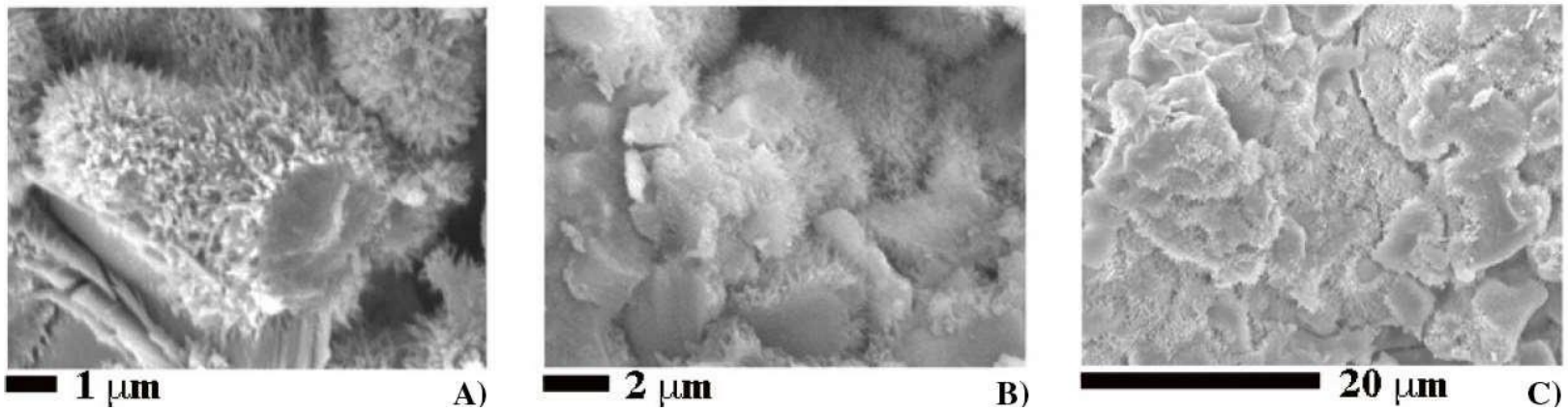
$$2(3CaO.SiO_2)+6H_2O \rightarrow 3\ CaO.2SiO.3H_2O + 3Ca(OH)_2$$

OR

$$2C_3S+6H \rightarrow C_3S_2H_3 + 3Ca(OH)_2$$

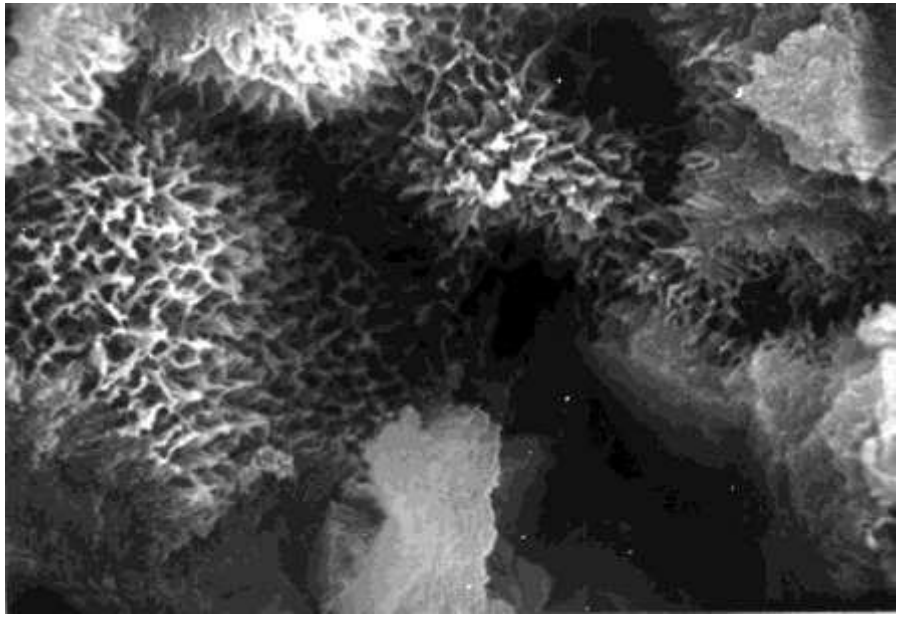
- Tri-calcium Silicate hydrates rapidly and forms earlier strength of the concrete .
- C_3S produce more amount heat during hydration process.
- Cement with more content of C_3S is better cold weather concreting.

HYDRATION PRODUCTS

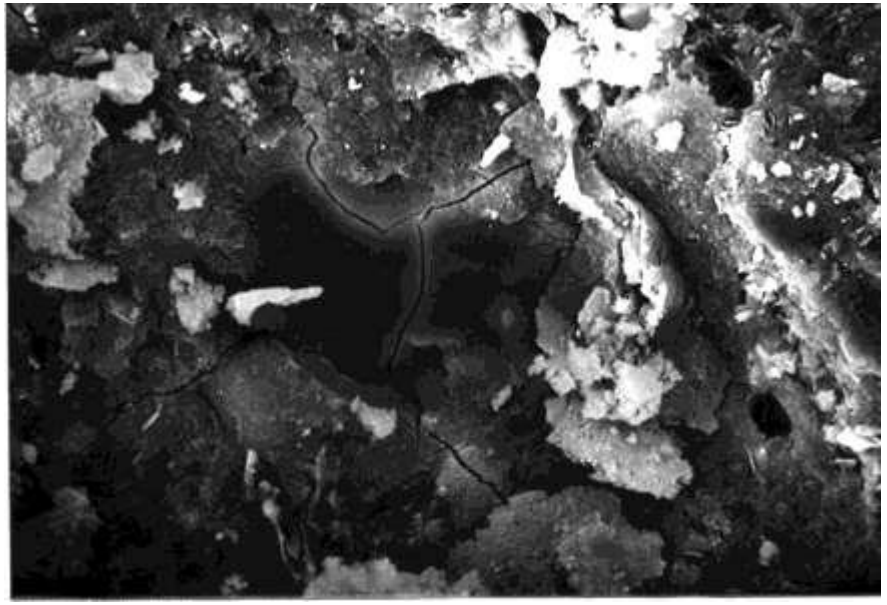


SEM micrographs of fractured C_3S pastes (w/c = 0.4) in pure water at (*A*) 7 days, (B) 13 days, (*C*) 1 month of hydration

HYDRATION PRODUCTS



Early Porous C-S-H gel (Eternite shingle, 2100x)



Late dense C-S-H gel (Eternite shingle, 800x)

HYDRATION PROCESS-C_2S

Calcium Silicate Hydrates

$$2(2CaO.SiO_2)+4H_2O \rightarrow 3CaO.2SiO.3H_2O + 3Ca(OH)_2$$

OR

$$2C_2S+4H \rightarrow C_3S_2H_3 + Ca(OH)_2$$

- Di-calcium Silicate hydrates slowly and it is responsible for progressive strength of the concrete.
- C_3S produce less amount heat during hydration process.
- Responsible for major strength of concrete.

HYDRATION PROCESS- $CA(OH)_2$

- The other products of hydration of C_3S and C_2S is Calcium Hydroxide.

- Lack of durability of concrete, is on account of the presence of calcium Hydroxide.

- $Ca(OH)_2$ reacts with sulphates presents in soil or water to form Calcium Sulphate which further reacts with C3A and causes the deterioration of concrete. This reaction is known as the sulphate attack.

HYDRATION PROCESS- $CA(OH)_2$

- The use of blending materials such as fly ash ,silica fume& other pozzolanic materials are steps to over come the sulphate attack.

Pozzolan + Calcium hydroxide+ water → C-S-H (Gel)

- Only advantage is that Ca(OH)2,being alkaline in nature maintain pH value around 13 in the concrete which resists the corrosion of reinforcements.

HYDRATION PROCESS-CALCIUM ALUMINATE HYDRATES (C3A&C4AF)

HYDRATION PROCESS-C_3A

Calcium Aluminate Hydrates

$3CaO.Al_2O_3+H_2O \rightarrow 3CaO.Al_2O_3.6H_2O$

OR

$C_3A + H \rightarrow C_3AH_6$

- Tri-calcium Aluminate hydrates rapidly about 225° C.
- C_3A produce large amount heat during hydration process.
- Hydrated aluminates do not contributes anything to the strength of concrete, but it will be give durability of concrete

HYDRATION PROCESS-C_4AF

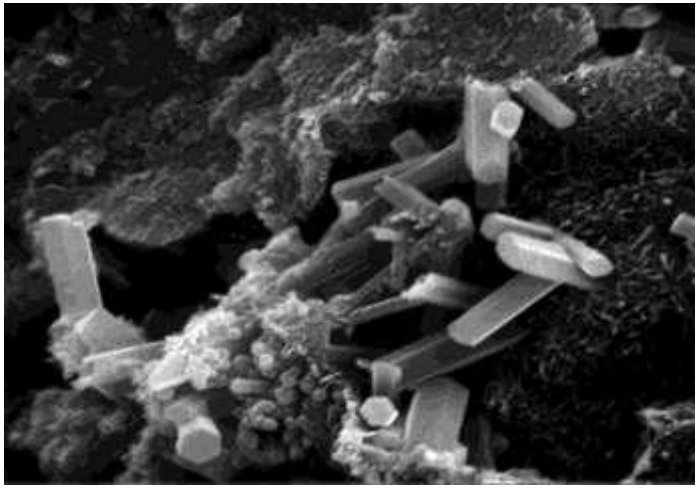
$4CaO.Al_2O_3.Fe_2O_3 + H_2O \rightarrow 3CaO.Al_2O_3.6H_2O + CaO.Fe_2O_3.H_2O$

OR

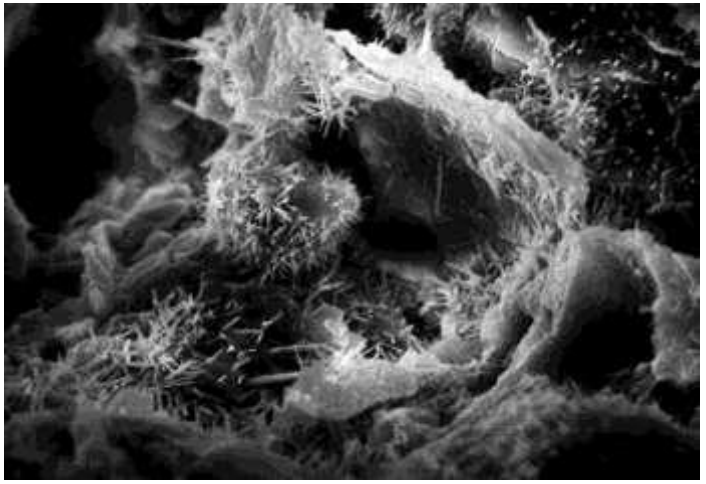
$C_4AF + H \rightarrow C_3AH_6$

- The hydrates of C4AF show a comparatively higher resistance to the attack of sulphates.

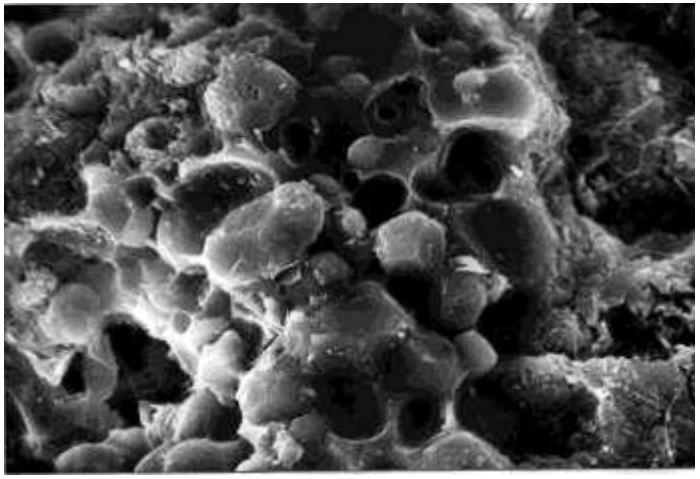
HYDRATION PRODUCTS



Lathshaped AFm (sulfatefree) crystals (3900x)



Ettringite (sulfatefree) crystals (1500x)



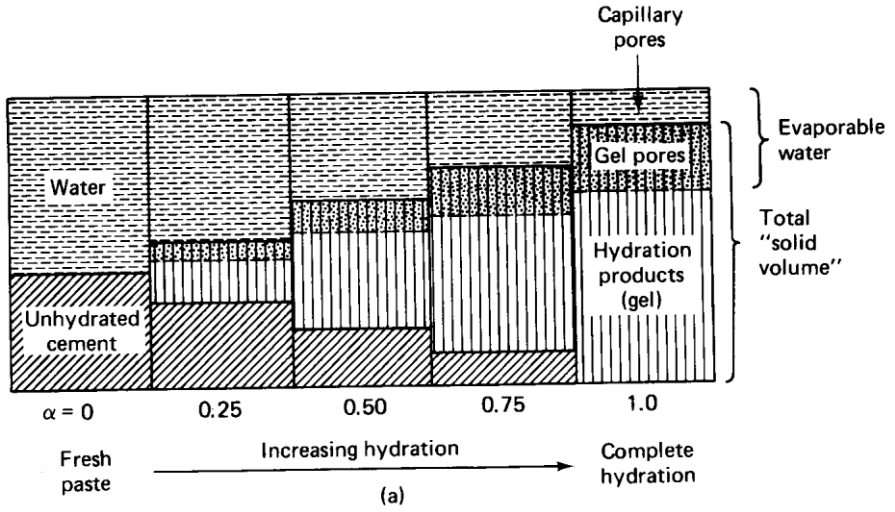
Hydrogarnets (1500x)

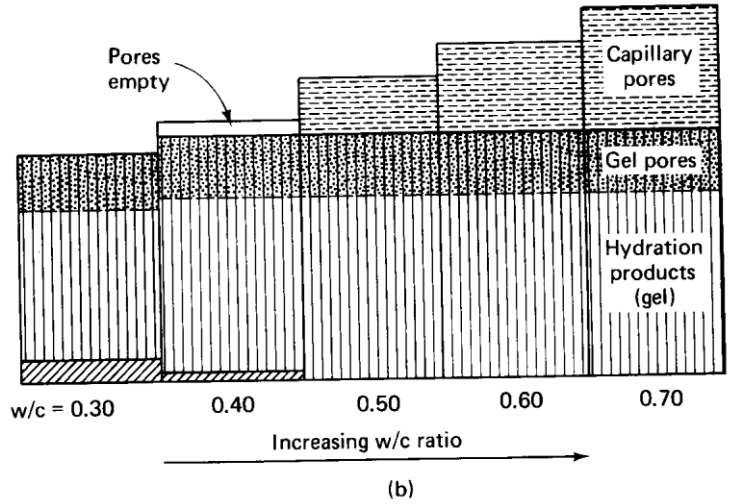
All images are from fiber concrete samples.

Volume relationships

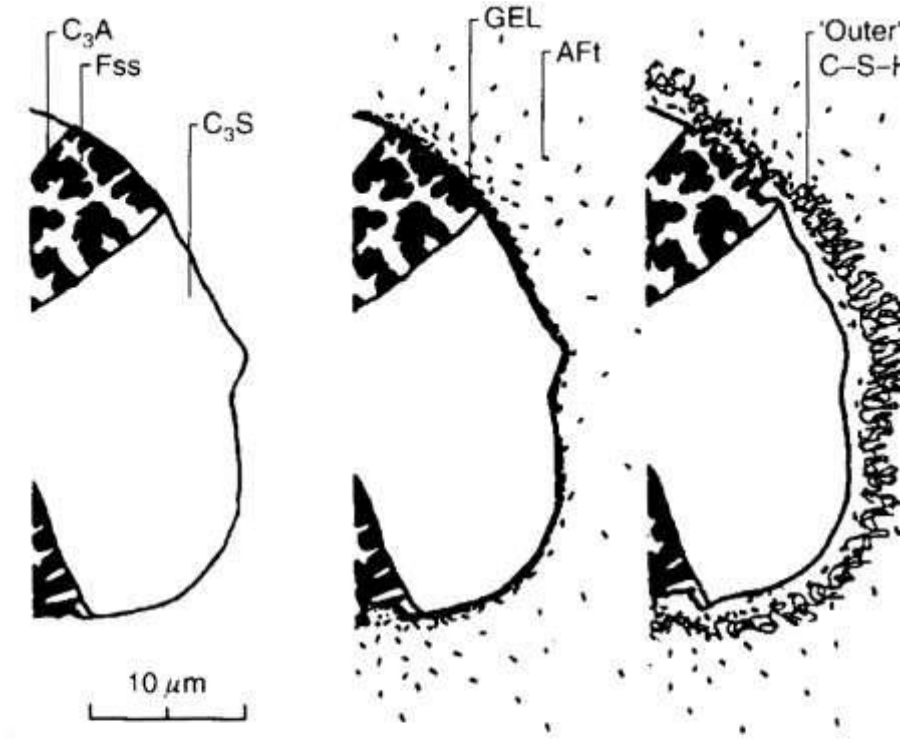
w/c is 0.5 for (a)

α is 1.0 for (b)





Hydration mechanisms in Portland cement

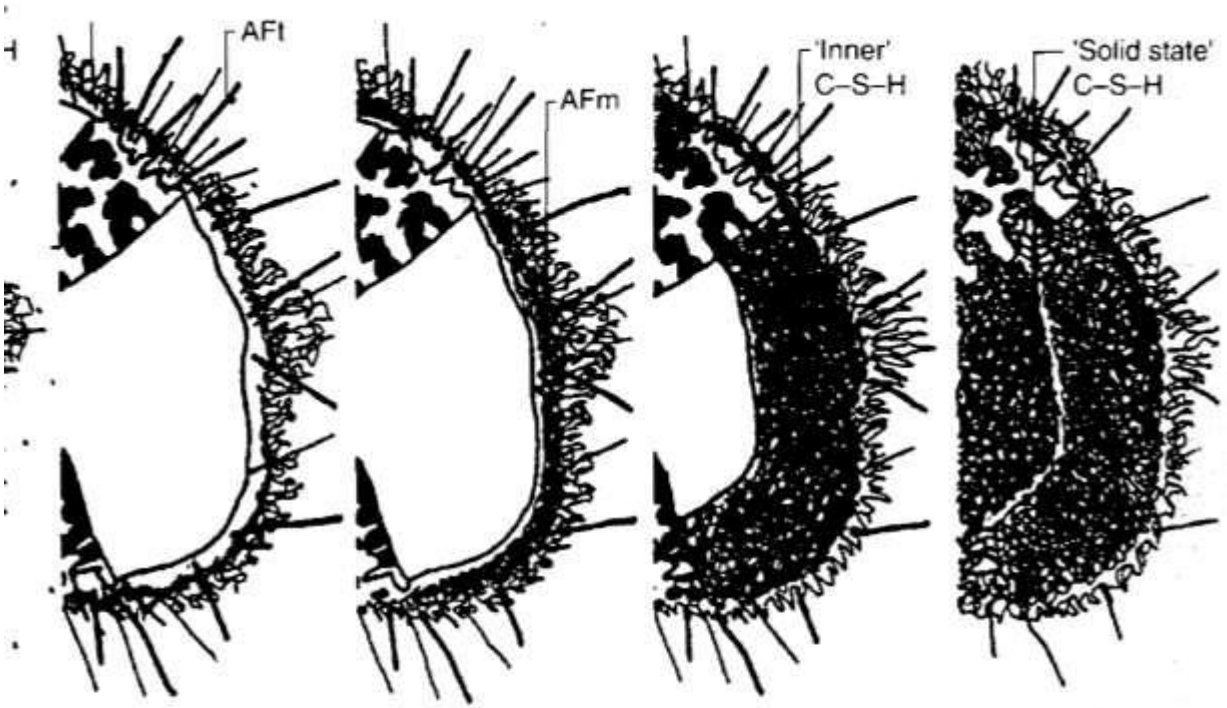


(a) Unhydrated section of poly-mineralic grain (Scale of interstitial phase is slightly exaggerated)

(b) –10 min Some C_3A (and/or Fss) reacts with calcium sulfate in solution. Amorphous, aluminate rich gel forms on the surface and short AFt rods nucleate at edge of gel and in solution

(c) –10 h Reaction of C_3S to produce 'outer' product C–S–H on AFt rod net-work leaving –1 μm between grain surface and hydrated shell

Hydration mechanisms in Portland cement



(d) –18 h
Secondary hydration of C_3A (and/or Fss) producing long rods of AFt. C–S–H 'inner' product starts to form on inside of shell from continuing hydration of C_3S

(e) 1–3 days
C_3A reacts with any AFt inside shell forming hexagonal plates of AFm. Continuing formation of 'inner' product reduces separation of anhydrous grain and hydrated shell

(f) –14 days
Sufficient 'inner' C–S–H has formed to fill in the space between grain and shell. The 'outer' C–S–H has become more fibrous

(g) – years
The remaining anhydrous material reacts by a slow solid state mechanism, to form additional 'inner' product C–S–H. The ferrite phase appears to remain unreacted

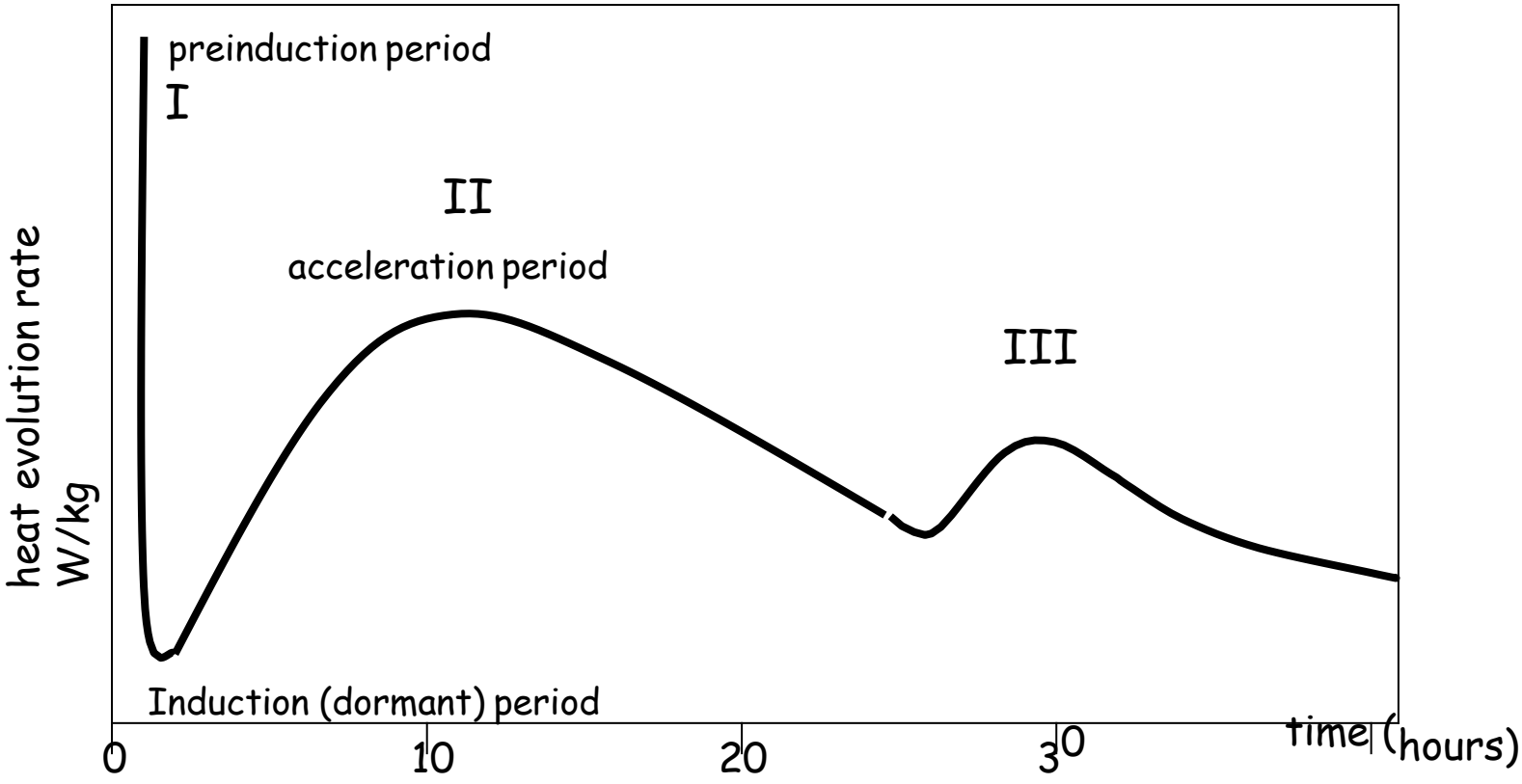
HEAT HYDRATION

- The Hydration of cement with water is exothermic.
- Exothermic-An exothermic reaction is a chemical reaction that releases by light or heat.
- The liberation of heat is called heat of hydration.

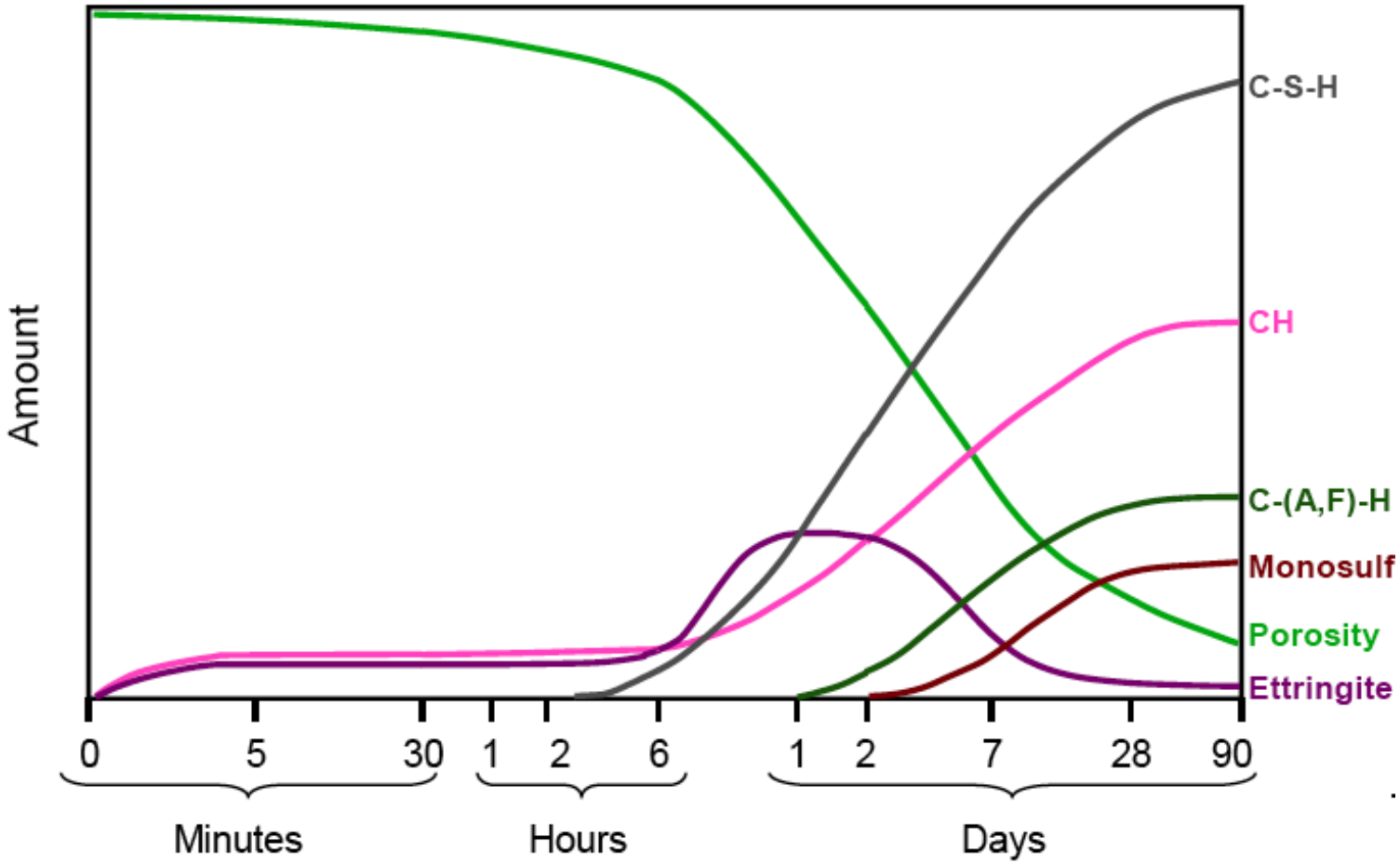
HEAT HYDRATION

- Study and control of hydration becomes important in mass construction(like dams.etc.,)
- Heat evolution due to the reaction of solution of aluminates(C_3S) and sulphates.(C_3A)
- Initial heat evolution ceases quickly when the solubility of aluminates is depressed by gypsum.
- The next heat evolution is on account of formation of ettringite and reaction of C_3S.
- Ettringite-Calcium Aluminate tri-sulphate Hydrates($C_6AS_3H_{32}$).

HEAT EVOLUTION DURING PORTLAND CEMENT HYDRATION

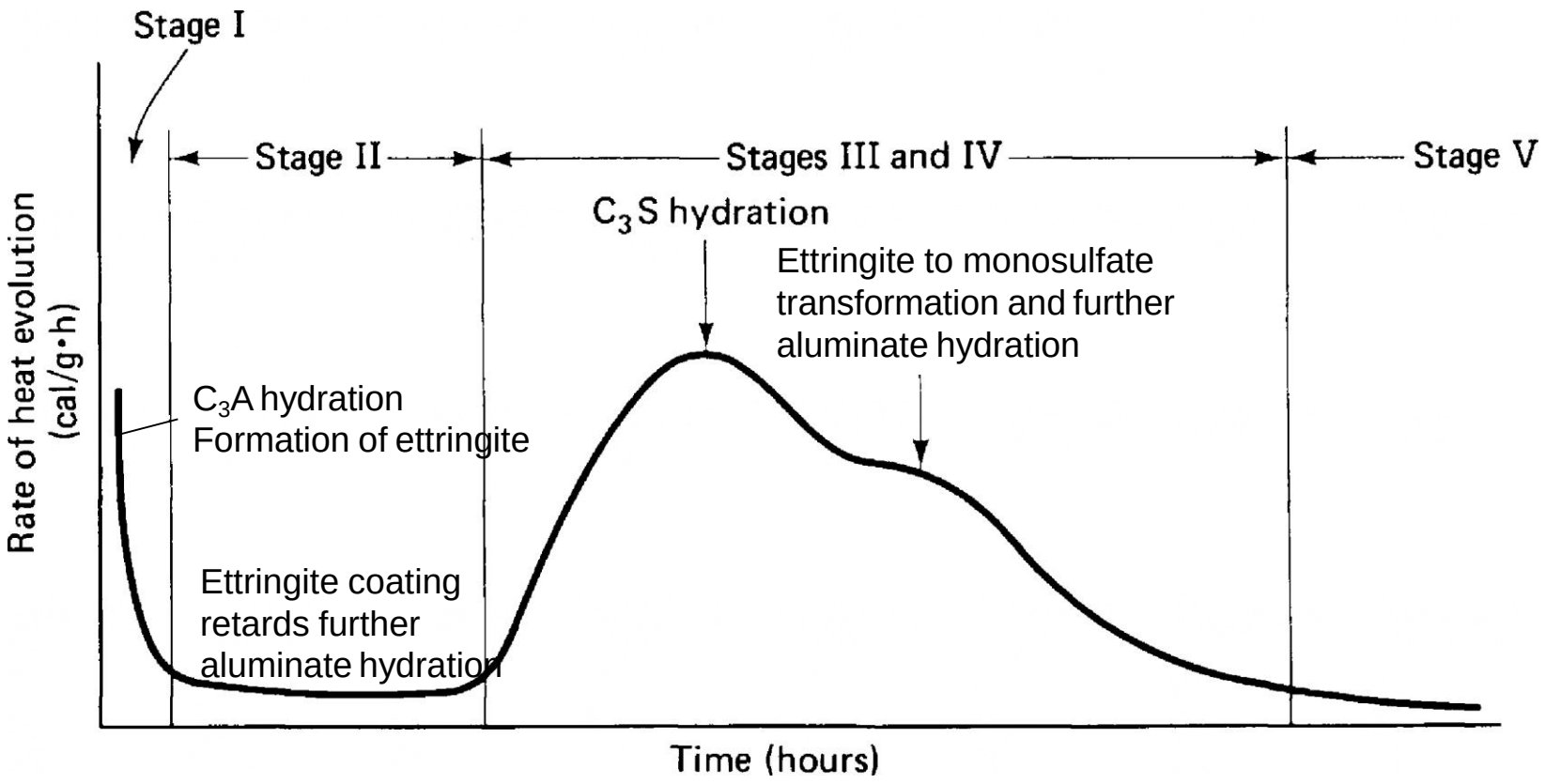


TEMPORAL EVOLUTION OF THE PRODUCTS



Temporal evolution of the hydration hydration products (Kurtis,)

HEAT EVOLUTION AND HYDRATION REACTIONS I



Relationship between reactions and heat evolution

Factors affecting Hydration

- Chemical Composition of Cement
- Cement Type
- Sulfate Content
- Fineness of Cement
- Water Cement Ratio
- Curing Temperature
- Effect of Admixtures

TYPES OF CEMENT:

1. Ordinary Portland Cement
2. Rapid Hardening Cement (or) High Early Strength cement
3. Extra Rapid Hardening Cement
4. Sulphate Resisting Cement
5. Quick Setting Cement
6. Low Heat Cement
7. Portland Pozzolana Cement
8. Portland Slag Cement
9. High Alumina Cement
10. Air Entraining Cement
11. Supersulphated Cement
12. Masonry Cement
13. Expansive Cement
14. Colored Cement
15. White Cement

(1) ORDINARY PORTLAND CEMENT:

- It is called Portland cement because on hardening (setting) its colour resembles to rocks near Portland in England. It was first of all introduced in 1824 by Joseph Asp din, England.

- Most important type

- Classified into three grades, namely 33 grade, 43 grade and 53 grade.

(2) RAPID HARDENING CEMENT:

- Also known as early gain in strength of cement. This cement contains more %age of C3S and less %age of C2S, high proportion of C3S will impart quicker hydration
- The high strength at early stage is due to finer grinding, as fineness of cement will expose greater surface area for the action of water.
- The strength obtained by this cement in 03 days is same as obtained by O.P.C in 7 days.
- Initial and final setting times are same as OPC.ie. 30mins and 10 hrs.And soundness test by Le-Chatielier is 10mm and Autoclave is 0.8%.
- Greater lime content than OPC

(3) EXTRA RAPID HARDENING CEMENT:

- It is obtained by intergrinding Cacl2 with rapid hardening cement.
- Addition of Cacl2 should not exceed 2% by weight of the rapid hardening cement.
- Concrete made by using this cement should be transported, placed, compacted & finished within about 20 minutes.
- Strength is higher than 25% than that of rapid hardening cement at 1 or 2 days.

(4) SULPHATE RESISTING CEMENT:

- It is modified form of O.P.C and is specially manufactured to resist the sulphates.
- This cement contains a low %age of C3A and high %age of C3S
- This cement requires longer period of curing.
- It develops strength slowly, but ultimately it is as strong as O.P.C.

(5) QUICK SETTING CEMENT:

- This cement is manufactured by adding small %age of aluminum sulphate (Al2SO4) which accelerates the setting action.
- Gypsum content is reduced.
- Sets faster than OPC.
- Initial setting time is 5 minutes. Final setting time is 30 minutes.

(6) LOW HEAT CEMENT:

- Low percentage of tri-calcium aluminates (C3A) and silicate (C3S) and high %age of di-calcium silicate (C2S) to keep heat generation low.
- Very slow rate of developing strength as rate of C3S Content is low.
- Heat evolved @ 7 days-66 cal/g and 28 days-75 cal/g
- initial set time-1 hr, final set time-10 hrs
- Better resistance to chemical attack than OPC.

(7) Portland Pozzolana Cement:

- OPC clinker and Pozzolana (Calcined Clay, Surkhi and Fly ash) ground together.

Produces less heat of hydration and offers great resistance to attacks of Sulphates.

- Used in marine works and mass concreting.
- Ultimate strength is more than OPC.
- Low shrinkage on drying
- Water tightness.

(8) Portland Slag Cement:

- Produced by mixing Portland cement clinker, gypsum and granulated blast furnace slag which shall not exceed 65%
- blackish grey in color.
- Lesser heat of hydration.
- Suitable for marine works, mass concreting.
- Offers good resistance to the attack of sulphate.

(9) HIGH ALUMINA CEMENT:

Different from OPC

Characterised by its dark colour, high heat of hydration and resistance to chemical attack.

Initial setting time of 4 hrs and final setting time of 5 hrs.

Raw materials used are limestone and bauxite

(10) AIR ENTRAINING CEMENT:

- OPC with small quantity of air entraining materials (oils, fats, fatty acids) ground together.
- Air is entrained in the form of tiny air bubbles which enhances workability and reduces seggregation and bleeding.
- It increases sulphate water resistance of concrete.

(11) Supersulphated Cement:

•Ground blast furnace slag + OPC +CASO4.

Heat of hydration which is considerably lower.

- It is also resistant to Sulphate attack.
- Used in a) Marine Structures, b) Mass concrete works

(12) **Masonry Cement:**

- Unlike ordinary cement, it is more plastic.
- Made by mixing hydrated lime, crushed stone, granulated slag or highly colloidal clays are mixed with it.
- Addition of above mentioned materials reduces the strength of cement.

(13) Expansive Cement:

- The main difference in this cement is the increase in volume that occurs when it settles.
- Used to neutralize shrinkage of concrete made from ordinary cement so as to eliminate cracks. A small percentage of this cement with concrete will not let it crack. It is specially desirable for hydraulic structures.
- In repair work, it is essential that the new concrete should be tight fitting in the old concrete. This can be done by using this cement

(14) Colored Cement:

- Suitable pigments used to impart desired color.
- Pigments used should be durable under light, sun or weather.

(15) WHITE CEMENT:

- OPC with pure white color produced with white chalk or clay free from iron oxide.
- As iron oxide gives the grey colour to cement, it is therefore necessary for white cement to keep the content of iron oxide as low as possible.
- Instead of coal, oil fuel is used for burning.

Special Types of Cement

- Acid Resistance Cement
- Blast Furnace Cement
- Expanding Cement
- Coloured Cement
- High Alumina Cement
- Hydrophobic Cement
- Low Heat Cement
- Oil Well Cement
- IRS-T 40 Special Cement

ACID RESISTANCE CEMENT

- This Cement composed as following,

  1.Acid-Resistance Aggregate – Ex: Quartz

  2.Additive- Ex: Sodium Fluosilicate

  (Na_2 SiF_6)

  3.Aqueous Solution of sodium silicate or soluble glass.

- Na_2 SiF_6 - Accelerates the hardening and increase the resistance of cement to acid.
- Binding Material of acid resistance cement is soluble glass.

BLAST FURNACE CEMENT

- For this cement Slag obtained from blast furnace is used.
- The slag is waste product in the manufacturing process of pig iron and it contains.
- It contains alumina, lime and silica.
- This cement more or less same as Ordinary Portland Cement.
- Its strength in early days is less and hence is requires longer curing period.

EXPANDING CEMENT

- This cement produced by adding an expanding medium like Sulpo-aluminate.
- About 8-20 parts of the Sulpo-aluminate clinker are mixed with 100 parts of Portland cement and 15 parts of stabilizer.
- Use of expanding cement requires skill and experience.
- This cement used in Water retaining Structures and repairing damaged concrete surface.

COLOURED CEMENT

- Coloured cement will be obtained by mixing of pigments with ordinary Portland cement.
- The amount of coloring material may vary from 5 to 10 %.
- If this percentage exceeds 10 % the strength of cement is affected.

Sl .No	Pigment	Colour
1	Chromium Oxide	Green
2	Cobalt Imparts	Blue
3	Iron Oxide in different proportion	Brown, Red, Yellow
4	Manganese Dioxide	Black or Brown

- Coloured cement are widely used Finishing Interior exterior, Flooring and artificial marble.

HIGH ALUMINA CEMENT

- This cement produced by grinding clinkers formed by bauxite and lime. The bauxite is an aluminum Oxide(Al_2O_3).
- In this cement alumina content should not less then 32%.
- Ratio of weight of alumina to the lime should not be between 0.85 to 1.30.
- It can stand high temperature.
- It sets quickly and attains higher ultimate strength short period.
- It is not suitable for mass construction as it evolves great heat.

HYDROPHOBIC CEMENT

- This type of cement contains admixtures which decrease the wetting ability of cement grains.
- The usual hydrophobic admixtures are acidol, naphthenesoap, oxidized petrolatum, Oletic acid and Stearic acid.
- Above substance formed water repellent film around each cement grains.
- This film protects them form the bad effect of moisture during storage and transportation.
- The film is broken out when the cement and aggregate are mixed together at the mixer exposing the cement particles to normal hydration.

Low Heat Cement

- Formation of cracks in large body of concrete due to heat of hydration.
- So, concrete technologists produce a kind of cement which produces less heat during hydration process.
- This cement will be used in mass construction.
- Low heat evolution is achieved by reducing the contents of C_3S C_3A.

Sl. No	Days	Heat produced by low heat cement (cal/gm)	Heat produced by OPC (cal/gm)	Cal-Calories 1 Calories = 4.2 joules
1	7	Not more than 65	89-90	
2	28	Not more than 75	90-100	

OIL WELL CEMENT

- Oil well are drilled through sedimentary rocks through a great depth of in search of oil.
- That oil or gas may escape through the space b/w steel casing and rock formation.
- This cement may resist corrosive condition from sulphur gases or water contains dissolved salts.
- It will be used in 175°C temperature and 1300 kg/m^{2} pressure.
- This cement obtained in two ways:
  i. Adjust cement compounds or adding Retarders.
  ii. Adding workability agents.

IRS-T 40 SPECIAL CEMENT

- This cement is manufactured as per specification laid down by ministry of Railways under IRS-T40: 1985.
- It is Very fine cement with high C_3S content for high early strength.
- It will be used to manufacture of concrete sleepers for Indian Railways.
- This cement also used in prestressed concrete elements, high rise buildings, high strength concrete.

CEMENT TEST

1.	**Normal Consistency of Cement**
2.	**Setting Time of Cement**
3.	**Specific Gravity of Cement**
4.	**Fineness Test of Cement by Sieve Analysis**
5.	**Compression Strength Test**
6.	**Soundness by Le-Chatelier's Method**

NORMAL CONSISTENCY OF CEMENT

The standard consistency of a cement paste is defined as that consistency which will permit the vicat plunger to penetrate to a point 5 to 7 mm from the bottom of the vicat mould. For finding out initial setting time, final setting time, soundness of cement and compressive strength of cement, it is necessary to fix the quantity of water to be mixed in cement in each case.

PROCEDURE:

1.Prepare a paste of weighed quantity of cement (300 grams) with a weighed quantity of potable or distilled water, starting with 26% water of 300g of cement.

2.Take care that the time of gauging is not less than 3 minutes, not more than 5 minutes and the gauging shall be completed before setting occurs.

3.The gauging time shall be counted from the time of adding the water to the dry cement until commencing to fill the mould.

4.Fill the vicatmould with this paste, the mould resting upon a non porous plate.

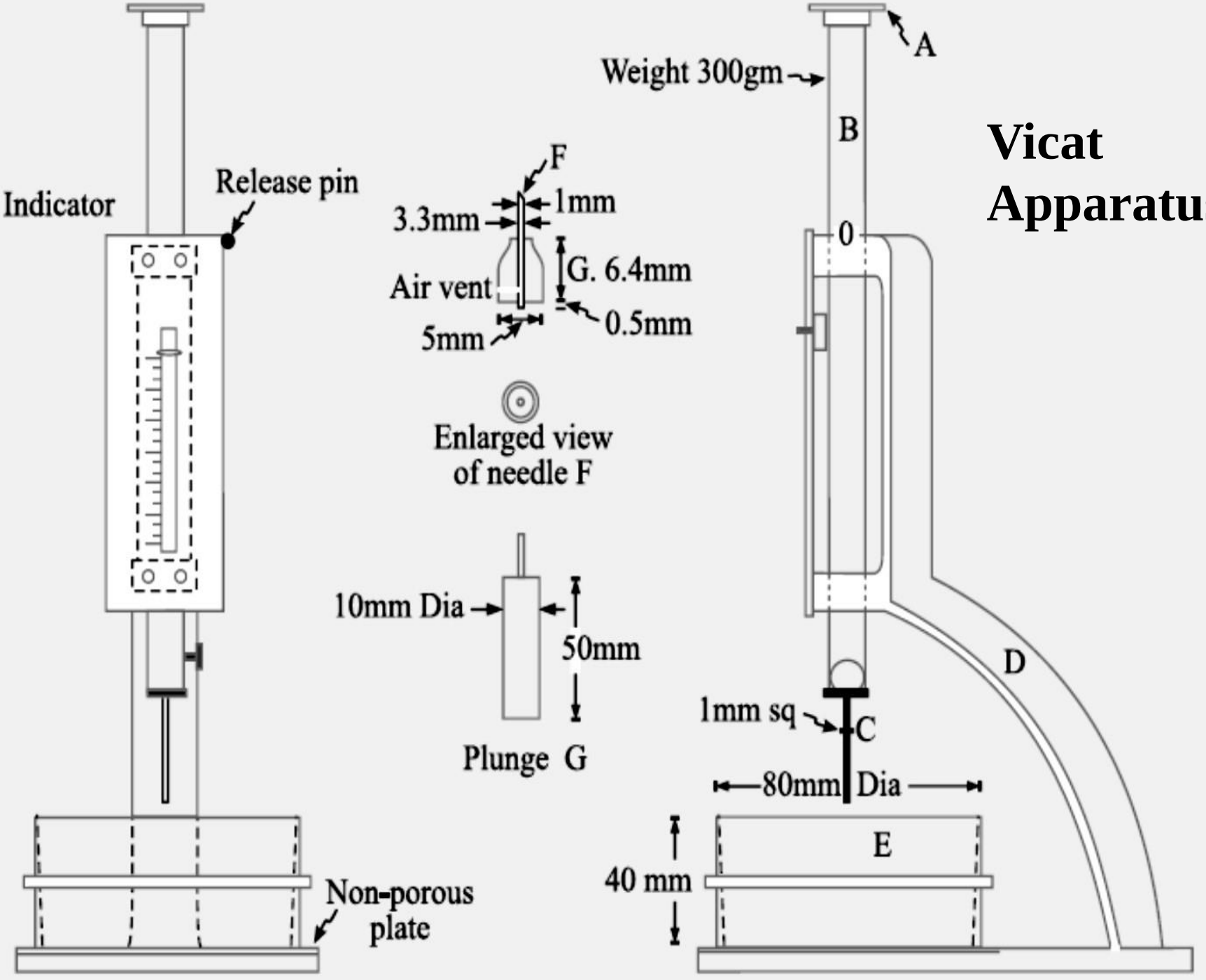
5.After completely filling the mould, trim off the surface of the paste, making it in level with the top of the mould. The mould may slightly be shaken to expel the air.

6.Place the test block with the mould, together with the non-porous resting plate, under the rod bearing the plunger (10mm diameter), lower the plunger gently to touch the surface of the test block and quickly release, allowing it to penetrate into the paste.

7.This operation shall be carried out immediately after filling the mould.

8.Prepare trial pastes with varying percentages of water and test as described above until the amount of water necessary for making the standard consistency as defined above is obtained

Vicat Apparatus




S%
P%

OBSERVATION TABLE

Trail No.	Weight of cement W1 (gm)	Percentage by water of dry Cement (%)	Amount of water added W2(ml)	Reading on scale mm	W_2 / W_1	Standard consistency(p)
1						
2						
3						
4						

CALCULATIONS:

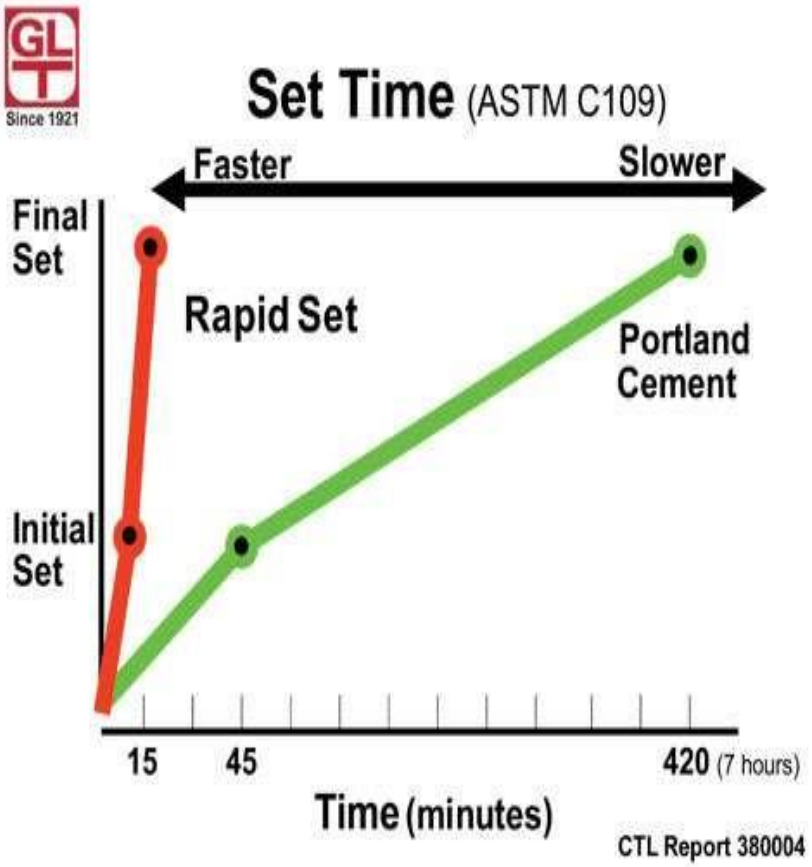
Weight of cement taken = W_1.

Weight of water added when the plunger has a penetration of 5 to 7 mm from the bottom of the mould = W_2

Percentage of water for standard consistency(p) = $(W_2 / W_1) \times 100$

Setting of Cement

- Setting-when water is mixed with cement, the action of changing from flux to solid state.
- During the next stage of hydration, paste gains strength and starts hardening.
- Initial setting time period-is the time required when the cement paste looses its plasticity. For OPC- 30mins
- Final setting time period- is the time required to achieve hardness. For OPC – 10hrs



SETTING AND HARDENING

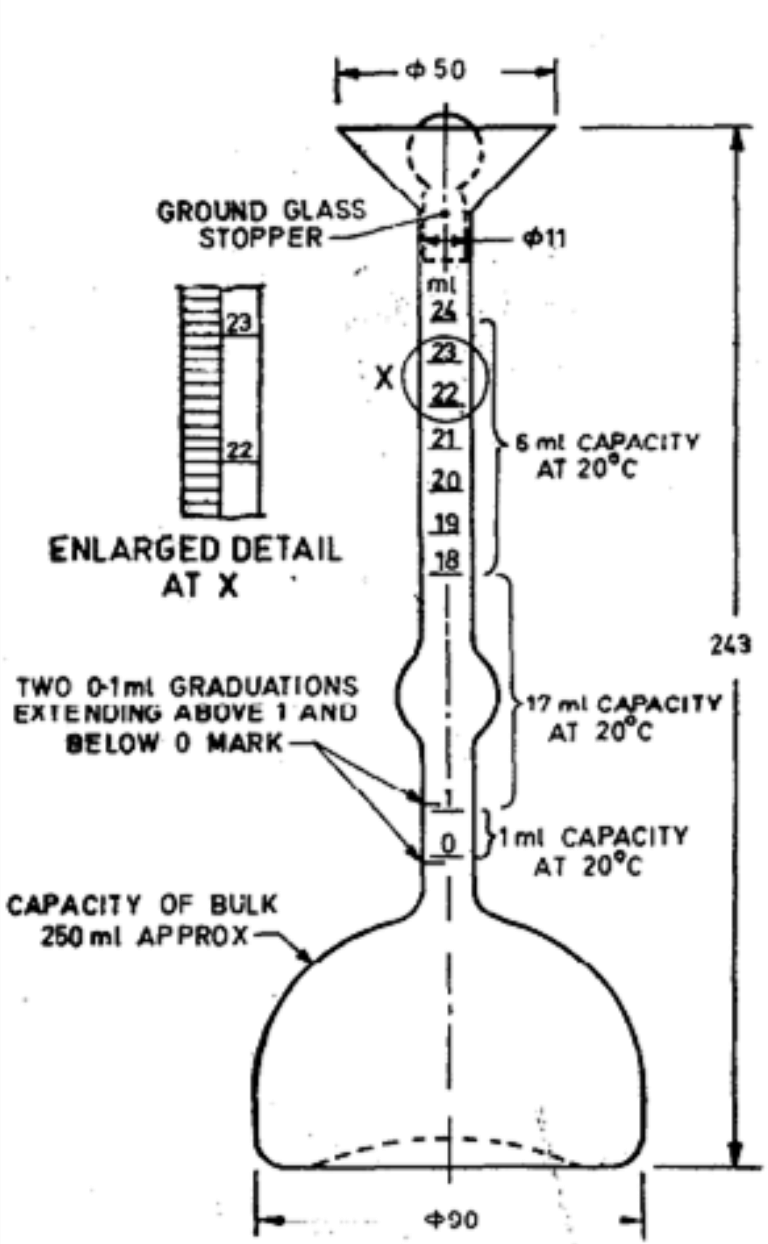
- Dormant Period – time from mixing cement with water and produce plastic workable paste. The paste remain virtually unchanged.
- Initial set – the paste begin to stiffen to such a degree, it becomes unworkable.
- Initial setting time – the time required for the paste to reach the stage.
- Final Set – the paste continues to stiffen until a stage is reached when it may be regarded as a rigid solid (complete solidification).

- Final setting time – the time required for the paste to reach the final set stage.

- Hardening – the process with time the hardened paste continues to harden and gain the strength.

SPECIFIC GRAVITY OF CEMENT





Procedure:

1.Dry the flask carefully and fill with kerosene or naphtha to a point on the stem between zero and 1 ml.

2.Record the level of the liquid in the flask as initial reading.

3.Put a weighted quantity of cement (about 60 g) into the flask so that level of kerosene rise to about 22 ml mark, care being taken to avoid splashing and to see that cement does not adhere to the sides of the above the liquid.

4.After putting all the cement to the flask, roll the flask gently in an inclined position to expel air until no further air bubble rise3s to the surface of the liquid.

5.Note down the new liquid level as final reading.

OBSERVATIONS:

$$\text{Specific gravity of Cement} = \frac{(W2 - W1) * (W4 - W1)}{((W4 - W1)-(W3-W2)) * (W5 - W1)}$$

Description of item	Trial 1	Trial 2
Weight of empty bottle(W1 g)		
Weight of bottle + Cement (W2 g)		
Weight of bottle + Cement + Kerosene(W3 g)		
Weight of bottle + Full Kerosene(W4 g)		
Weight of bottle + Full Water(W5 g)		

FINENESS TEST OF CEMENT BY SIEVE ANALYSIS

APPARATUS:

- IS: 90μ test sieve
- bottom pan
- weighing balance,
- brush

REFERENCE CODE: IS 4031 (PART1): 1988, IS460 (PART1): 1985

THEORY: The degree of fineness of cement is a measure of the mean size of the grains. The finer cement has quicker action with water and gains early strength without change in the ultimate strength. Finer cement is susceptible to shrinkage and cracking.

PROCEDURE:

1. Accurately weigh 100 gms of cement sample and place it over the test sieve. Gently breakdown the air set lumps if any with fingers.
2. Hold the sieve with pan in both hands and sieve with gentle wrist motion, in circular and vertical motion for a period of 10 to 15 minutes without any spilling of cement.
3. Place the cover on the sieve and remove the pan. Now tap the other side of the sieve with the handle of brush and clean the outer side of the sieve.
4. Empty the pan and fix it below the sieve and continue sieving as mentioned in the steps 2 and 3. Totally sieve for 15 minutes and weigh the residue (Left over the sieve).

OBSERVATIONS:

1. Weight of cement taken =.....
2. Weight of cement retained after sieving =.....
3. Type of cement =.....
4. Brand of cement=.....
5. Room temperature=.....

Percentage weight of Residue =

Weight of sample left on Sieve / Total Weight of Sample

COMPRESSIVE STRENGTH TEST

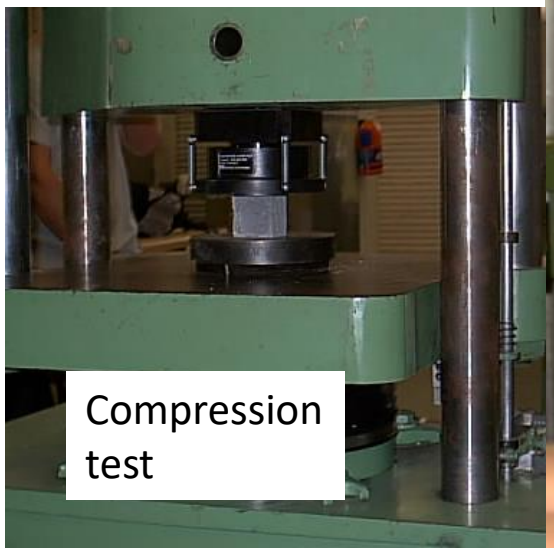
the compressive strength of cement mortars is determined in order to verify whether the cement conforms to IS specifications and whether it will be able to develop the required compressive strength of concrete. The average compressive strength of at least three mortar cubes (area of the face 50 cm2) composed of one part of cement and three parts of standard stand should satisfy IS code specifications.



Mold



Prepare sample



Compression test



Typical failure

OBSETRVATION:

Sr. No	Age of Cube	Dimensions of the specimen (mm)			Weight of Cement Cube (gms)	Cross-Sectional area (mm^2)	Crushing Load (N)	Compressive strength (MPa)	Average Compressive strength (MPa)
		L mm	B mm	H mm					
1	3 DAYS								
2									
3									
4	7 DAYS								
5									
6									
7	28 DAYS								
8									
9									

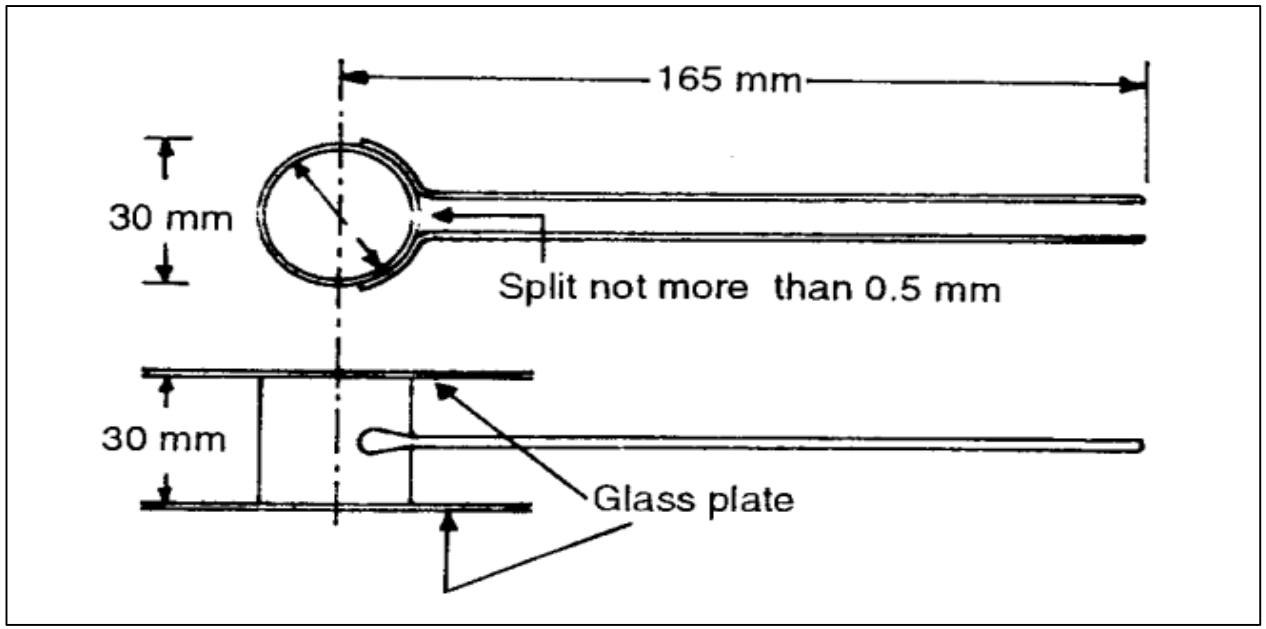
$$\text{COMPRESSIVE STRENGTH} = \frac{\text{CRUSHING LOAD}}{\text{CROSS SECTIONAL AREA}}$$

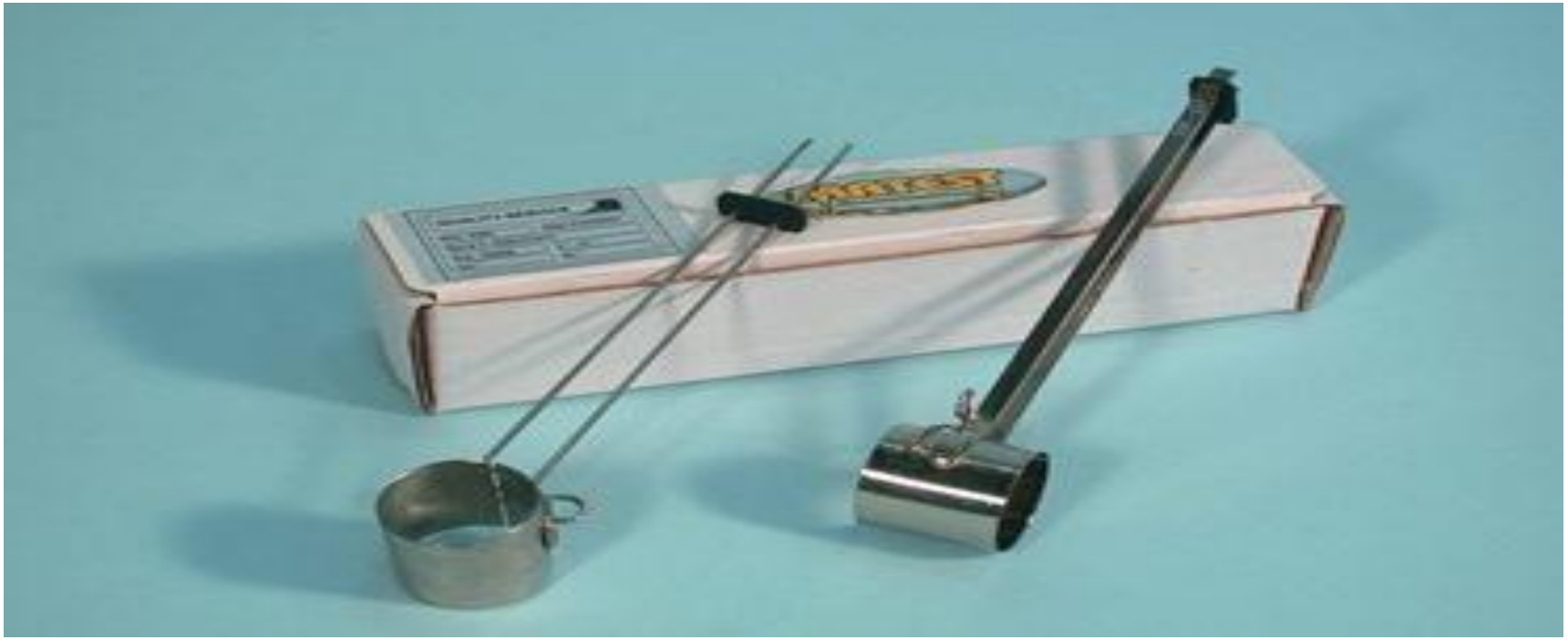
SOUNDNESS BY LE-CHATELIER'S METHOD

The ability of cement to maintain a constant volume is known as soundness of the cement. It is essential that the cement concrete shall not undergo appreciable change in volume after setting. Unsoundness produces cracks, distortion and disintegration there by giving passage to water and atmospheric gases which may have injurious effects on concrete and reinforcement. Soundness of cement is ensured by limiting the quantities of free lime, magnesia and sulphates as these compounds undergo a large change in volume.

Unsoundness in cement does not come to surface for a considerable period of time. Thus this test is designed to accelerate the hydration of free lime by the application of heat thus discovering the defects in a short time. Further, to minimise the shrinkage of cement paste, the test setup is kept immersed in water bath.

This test is carried out with the help of "Le Chatelier's apparatus" which consists of a small split cylinder of spring brass or other suitable metal of 0.5mm thickness forming a mould 30 mm internal diameter and 30mm high (Figure). On either side of the split mould are attached to indicators with pointed ends, the distance from these ends to the center of the cylinder being 165 mm. The mould shall be kept in good condition with the jaws not more than 50mm apart.


165 mm
30 mm
Split not more than 0.5 mm
30 mm
Glass plate



Procedure:

1. Weigh accurately 100 g of cement to the nearest 0.15 g and add to it 0.78 times the water required to give a paste of standard consistency (i.e. 0.78×P).
2. Place the lightly grease mould on a lightly grease glass sheet and fill it with cement paste, taking care to keep the edges of the mould gently together.
3. Cover the mould with another piece of lightly grease glass sheet, place a small weight on this covering glass sheet and immediately submerge the whole assembly in water at a temperature of 27 ± 20 CC.
4. Keep this assembly under water for 24 hrs. After this, take the mould out of water and measure the distance between two indicators. Submerge the mould again in the water.
5. Bring the water to boiling with the mould kept submerged, and keep it boiling for 25 to 30 minutes.
6. Remove the mould from the water allow it to cool and measure the distance between the indicator points.
7. The difference between these two measurements represents the expansion of the cement.
8. Repeat the whole procedures two more times each using fresh 100 g sample.

Observations:

Distance between pointers before boiling (D1) in mm =

Distance between pointers after boiling (D2) in mm =

Expansion of the cement = E1 = (D2- D1) in mm

THANK YOU